

Lab 1: Installation of windows 2019 (100 marks – 5%)

INFT 1103 – Fuad Mustapha



Madhvi Lashkaree (100997946)

Instructions

- This lab should be completed individually.
- All tasks in this lab are to be installed on your VMware workstation.
- Follow the steps taken in this Lab workbook and complete your installation. All screenshots and findings should be added to this manual.

OBJECTIVE

After completing this Assignment/Lab, you will be able to:

- Describe PKI Infrastructure, DHCP, DNS and DNS Server
- Install windows 2019 server Datacenter and change your server's name

Questions

1. Describe PKI infrastructure (research on the Internet)

Public Key Infrastructure (PKI) is a system that allows for secure electronic communication using digital certificates and public-key encryption.

The key components and functions of PKI:

- Public and private keys are used to encrypt data, with the former being shared publicly. A private key is kept secret by the owner and is used to decrypt data that has been encrypted with the public key.
- Digital certificates, based on the X.509 standard, bind public keys to the identification of persons or organizations. Certificate Authorities (CAs) are responsible for issuing these certificates.
- The Certificate Authority (CA) is the trusted third-party institution that issues, revokes, and manages certificates. It verifies the certificate requester's identity and digitally signs the certificate, which confirms the public key's legitimacy.

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- The Registration Authority (RA) functions as a middleman between users and the CA, verifying their identification before issuing a certificate. It does not issue certificates but rather simplifies the procedure for the CA.
- CAs maintain a Certificate Revocation List (CRL) with serial numbers of revoked certificates that are no longer trusted.
- The Public Key Cryptography Standards (PKCS) define formats for key generation, encryption methods, and certificate administration.

PKI working steps:

1. Each user gets a pair of keys - one public and one private.
2. The user asks for a digital certificate from a trusted authority.
3. The authority checks the user's identity.
4. The authority gives the user a digital certificate linking their identity to their public key.
5. Users can now send encrypted messages using each other's public keys, which can only be decrypted by the corresponding private keys.
6. Before trusting a certificate, users check if it's still valid and hasn't been revoked.

Examples of PKI:

Secure email, Encrypts web traffic (HTTPS), and User authentication.

2. Write short notes and explain the following (research on the Internet)

a. DHCP (Dynamic Host Configuration Protocol):

DHCP is a network management protocol that automatically assigns IP addresses and other network configuration settings (such as subnet mask, default gateway, and DNS servers) to networked devices. When a device connects to a network, it makes a request to the DHCP server. The server responds by assigning the device an available IP address from a preset range to avoid conflicts. This eliminates the need for manual IP address configuration, simplifies network management, and ensures the most efficient use of IP address resources.

b. DNS (Domain Name System):

DNS is a system that converts human-readable domain names such as www.google.com to machine-readable IP addresses such as 172.217.0.46. When a user enters a domain name into a browser, a DNS query is sent to a DNS server, which returns the appropriate IP address, allowing the user to visit the desired website. DNS allows consumers to easily access websites and internet services without having to remember complex IP addresses.

c. DNS Server:

An expert server that responds to queries converting domain names into IP addresses is known as a DNS server. An essential component of the internet's architecture, DNS servers enable effective domain-to-IP address resolution.

Types of DNS Servers:

- **Recursive DNS Server:** Responds to client queries (from browsers, for example) and does the necessary lookups to determine the proper IP address.
- **Authoritative DNS Server:** Responds to client queries (from browsers, for example) and does the necessary lookups to determine the proper IP address.

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Lab

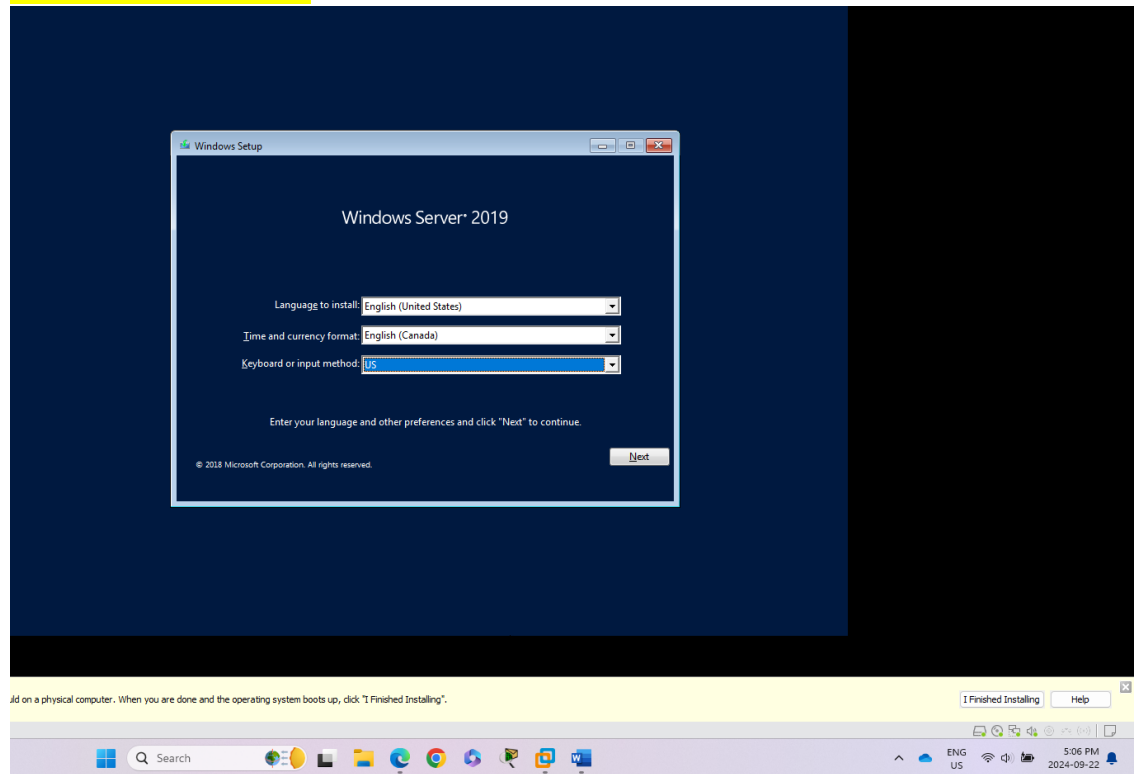
Install Windows 2019 server Datacenter (as an image on VMWare workstation)

Follow these steps to install **Windows Server 2019 Datacenter** as a virtual machine in **VMware Workstation**:

1. **Load the Windows Server 2019 ISO File**

- Open VMware Workstation and load the Windows Server 2019 Datacenter ISO.
- When the **Windows Setup** page appears, select your **language**, **time format**, and **keyboard layout**, then click **Next**.

[Attach Screenshot]



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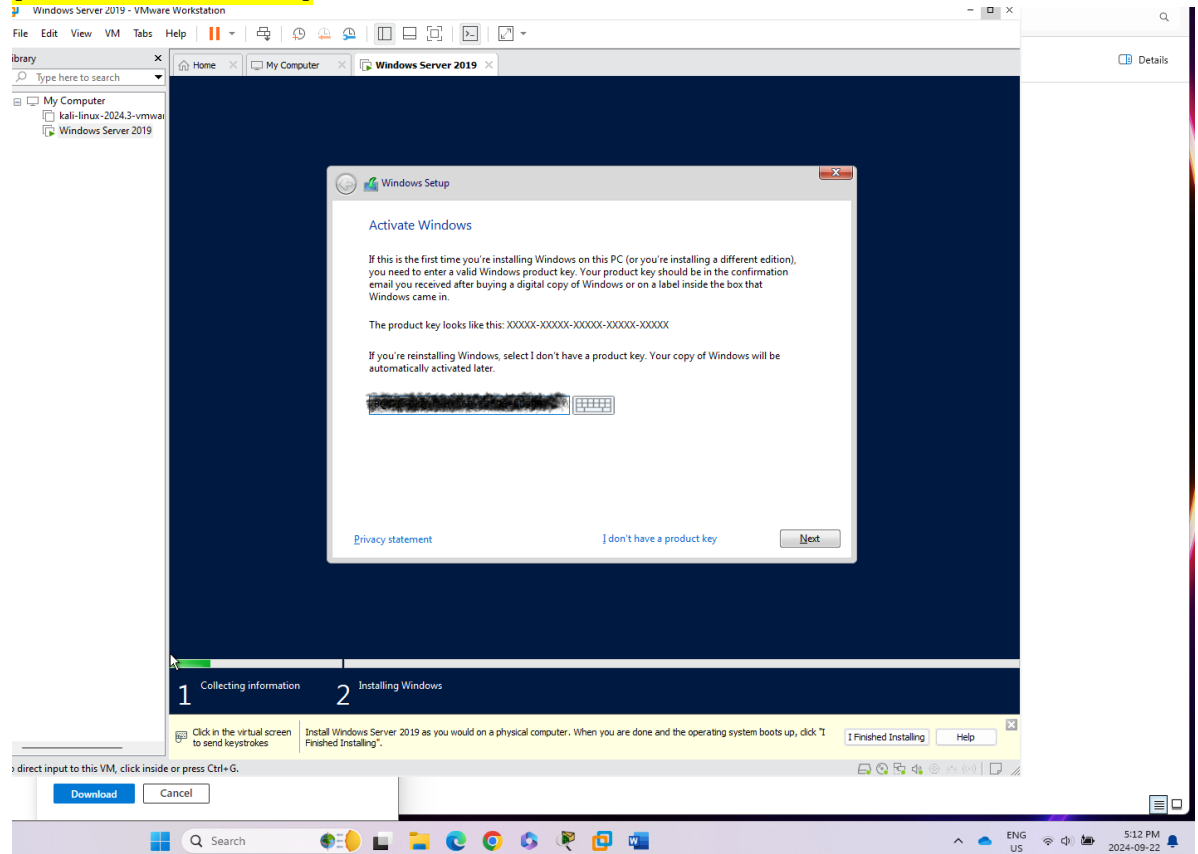
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2. Start the Installation Process

- Click on the **Install now** button to begin the installation.

[Attach Screenshot]



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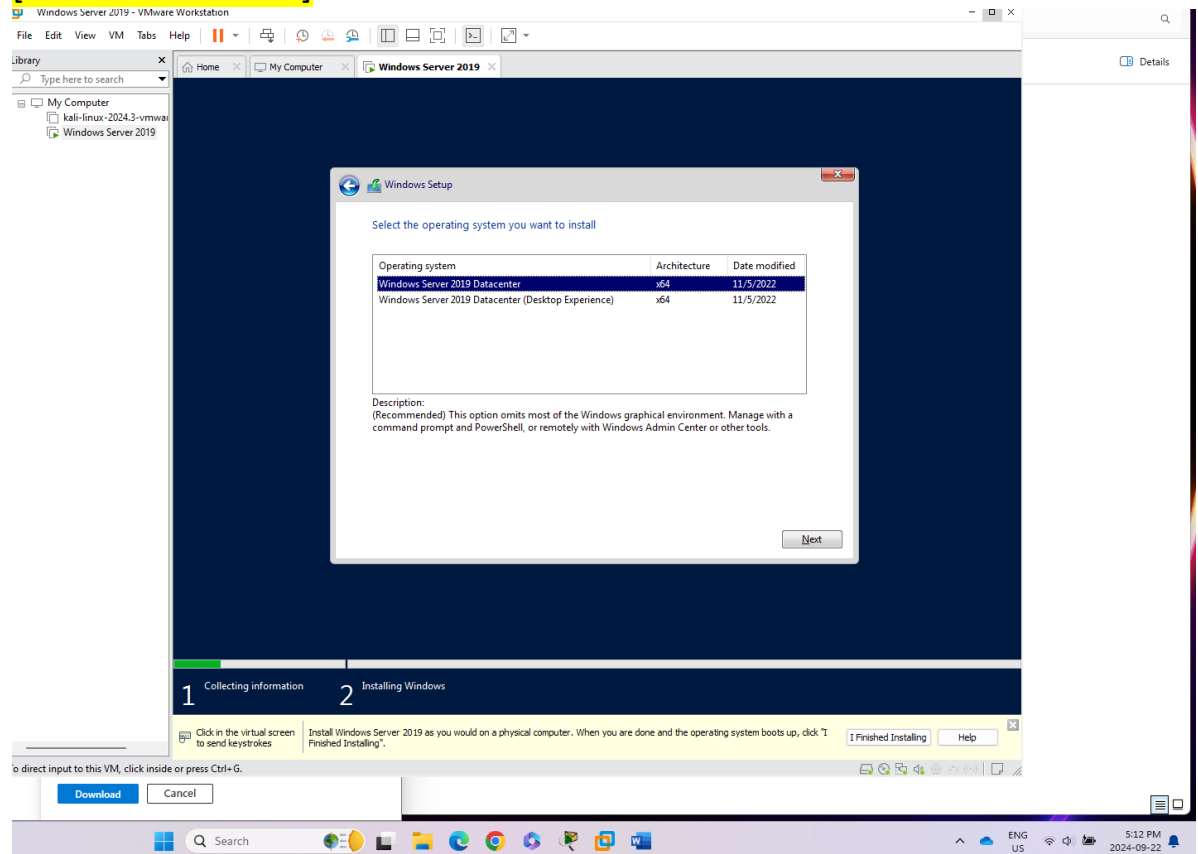
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3. Select Windows Server Version

- Choose **Windows Server 2019 Datacenter** to ensure you get the graphical interface.

[Attach Screenshot]



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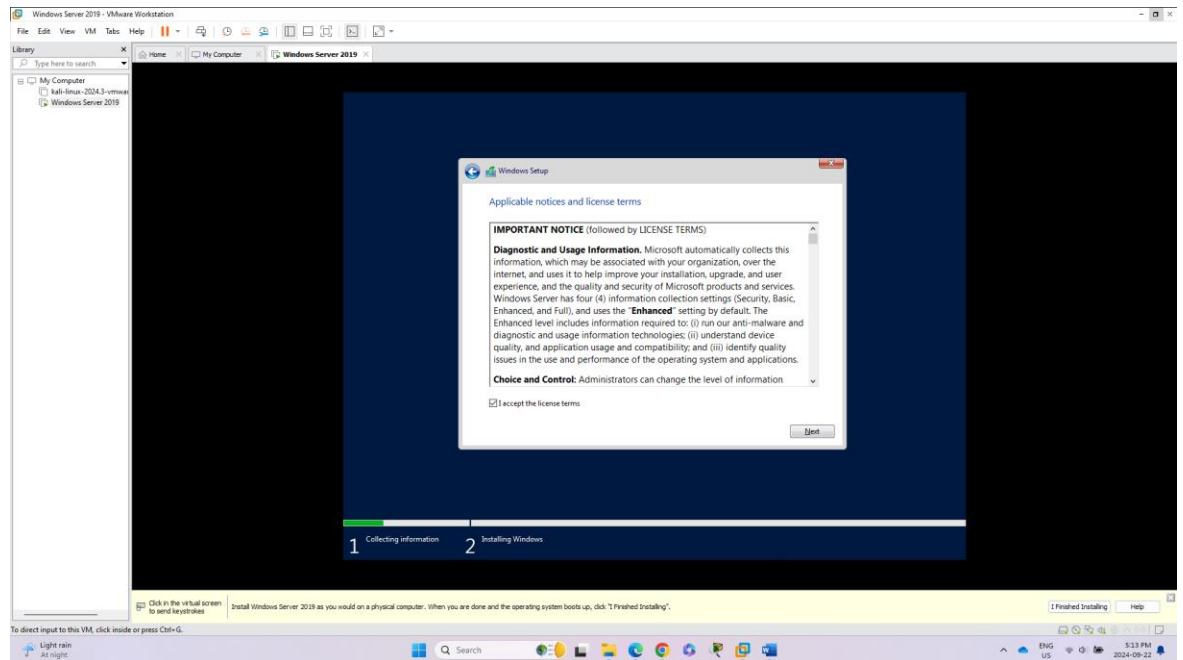
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4. Accept the License Terms

- Put a check mark in the box next to **I accept the license terms** and click **Next**.

[Attach Screenshot]



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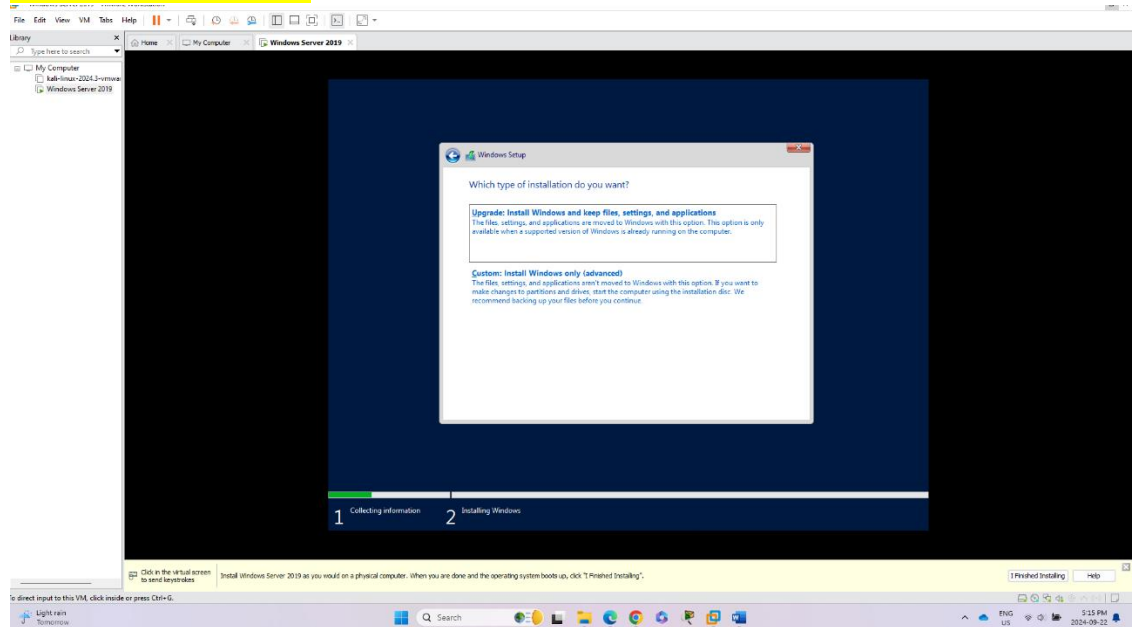
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5. Select Installation Type

- Select **Custom: Install Windows only (advanced)** for a clean installation.

[Attach Screenshot]



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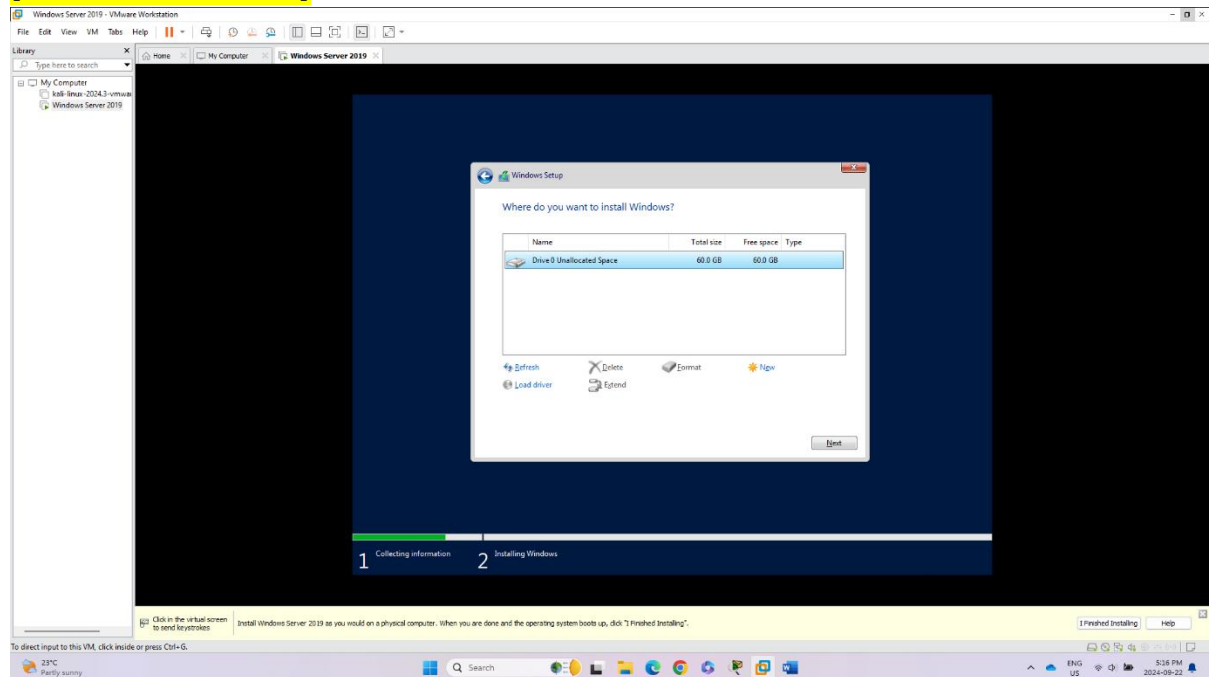
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6. Select and Partition the Hard Drive

- Choose the hard drive where you want to install Windows Server, then click the **New** button to create partitions.

[Attach Screenshot]



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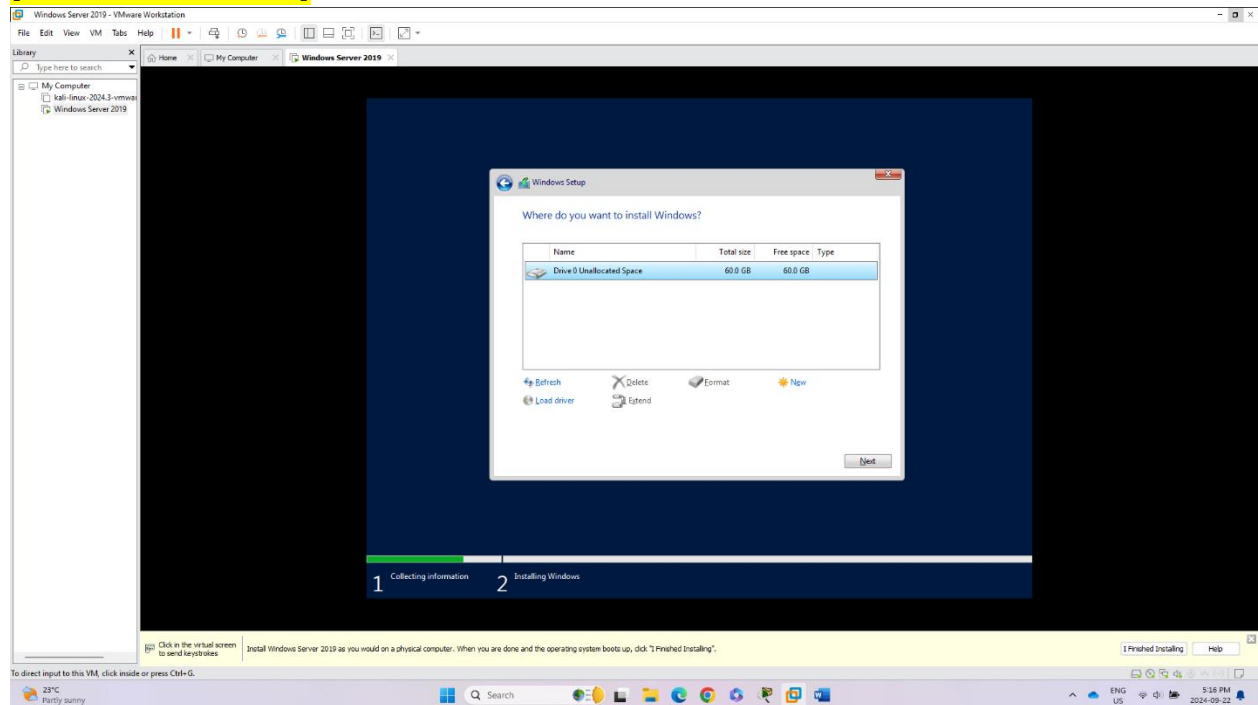
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7. Configure Partitions

- After clicking the **New** button, leave the default size and click **Apply**. A warning will appear asking for permission to create system partitions. Click **OK** and then hit **Next**.

[Attach Screenshot]



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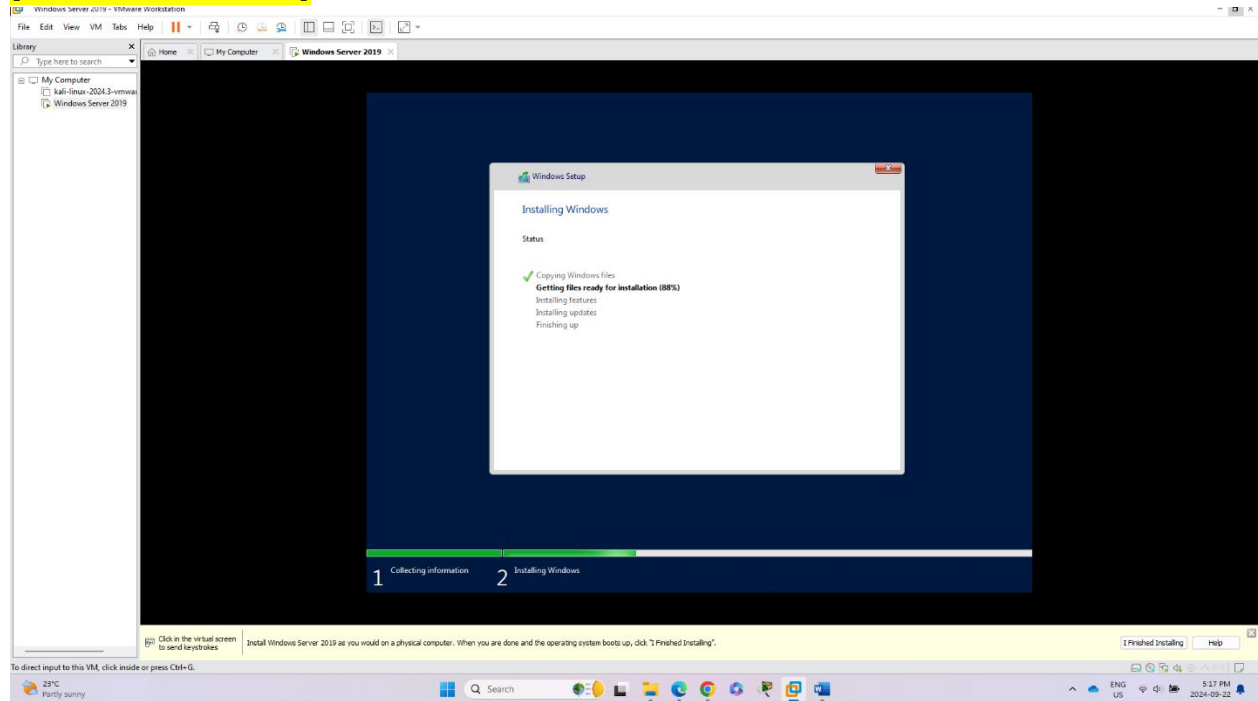
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8. Begin Installation

- Windows Server 2019 will now start installing. This process will take some time, and the system will restart several times. Grab a coffee and wait for it to finish.

[Attach Screenshot]



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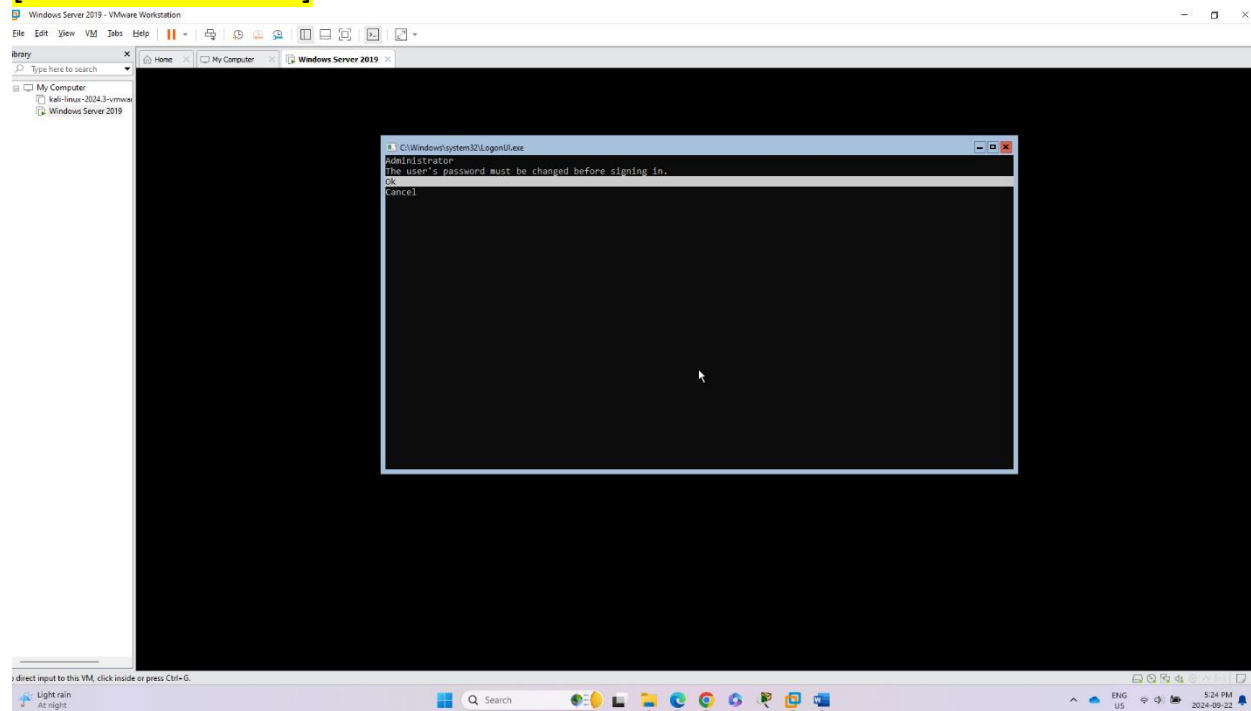
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9. Set Up Initial Configuration

- After installation, you'll be prompted to enter a password. Type a simple, yet secure password (e.g., **Password@123** or **P@ssword123**) twice, then click **Finish**.
- **Tip:** Use a password that meets Windows Server requirements (uppercase, lowercase, symbols, and numbers) and is easy to remember, especially since this is a lab environment.

[Attach Screenshot]



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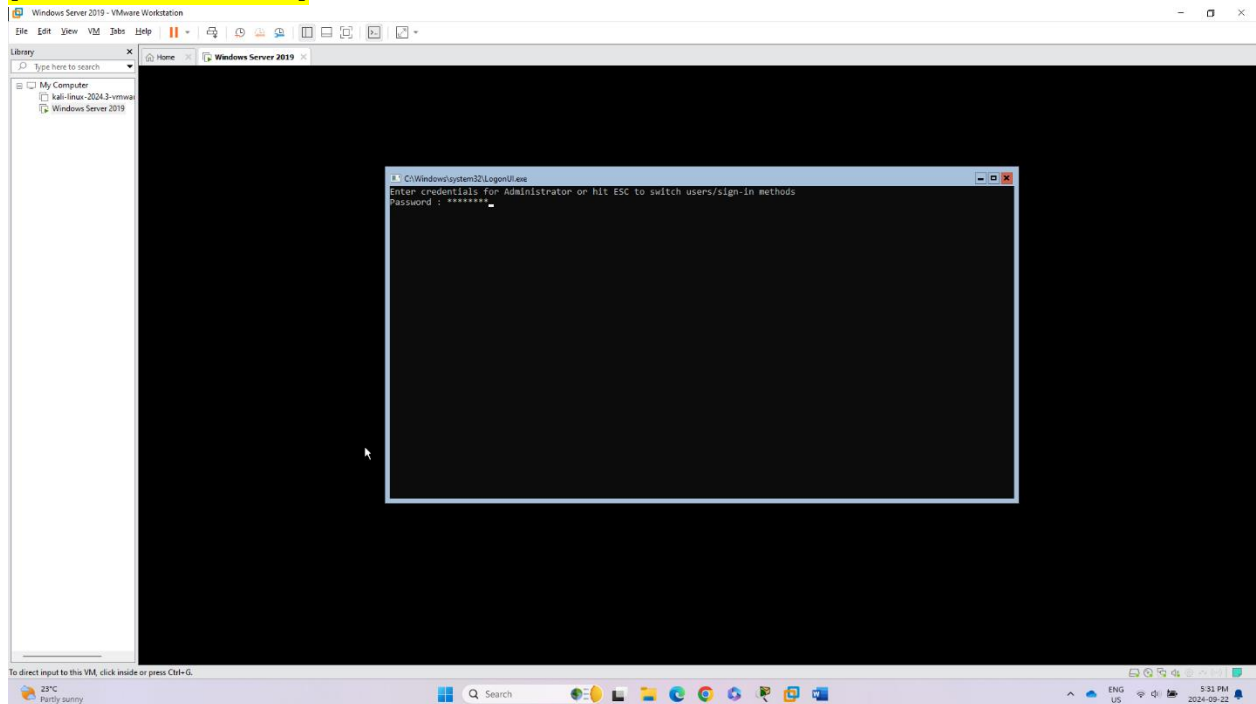
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10. Login to Your Server

- Press **Ctrl + Alt + Del** to bring up the login screen, then enter the password you created and press **Enter** to sign in.

[Attach Screenshot]



11. Rename Your Server

- Change the server's name to follow this format: **DC1-yourfirstname-last4digitsStudentID** (e.g., **DC1-Jane-5678**).
- To do this, open **Server Manager**, go to **Local Server**, click on the current **Computer Name**, and then select **Change** in the **System Properties** tab.
- Take three screenshots for the name change process:
 1. Screenshot of the current name.
 2. Screenshot after entering your new name and being prompted to restart.
 3. Screenshot of the new name after the restart.

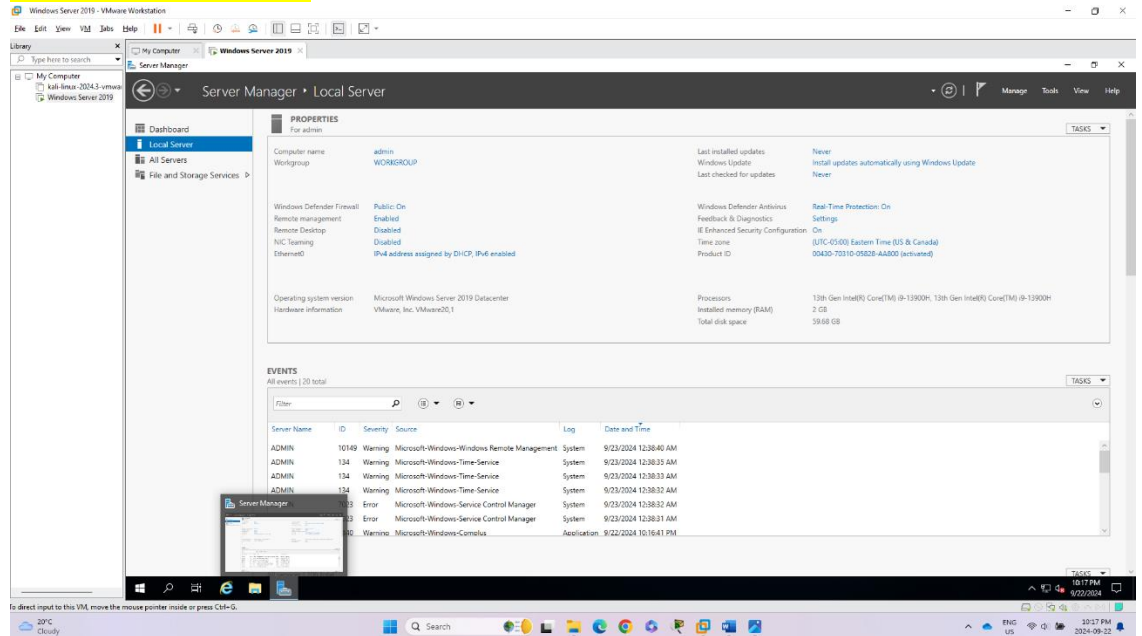
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INFT 1103 Operating System Security

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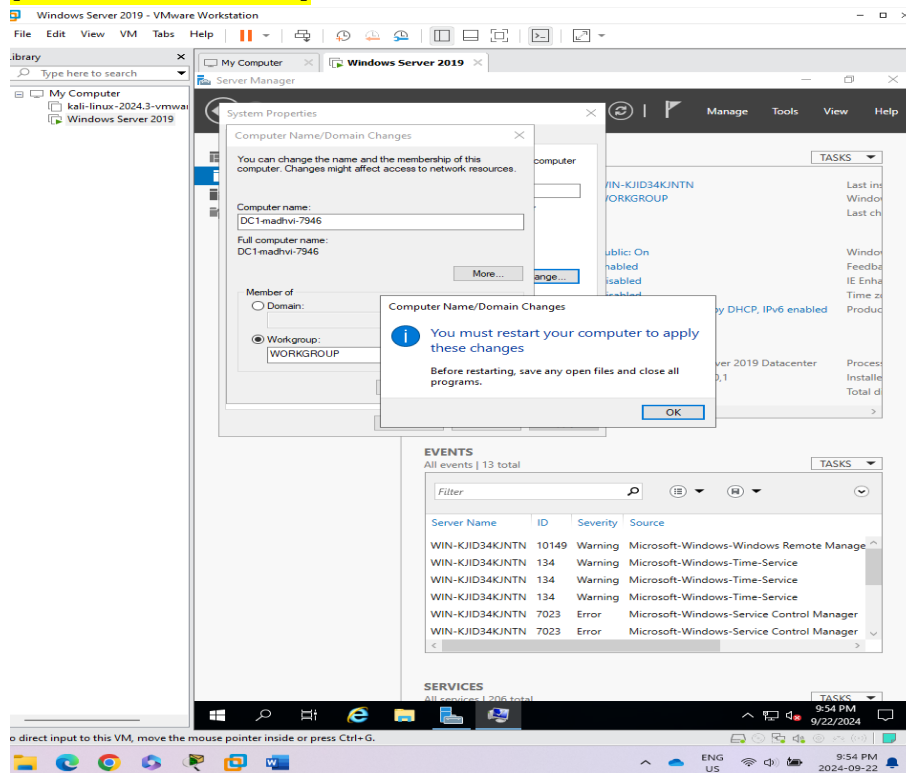
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[Attach Screenshot]



[Attach Screenshot]



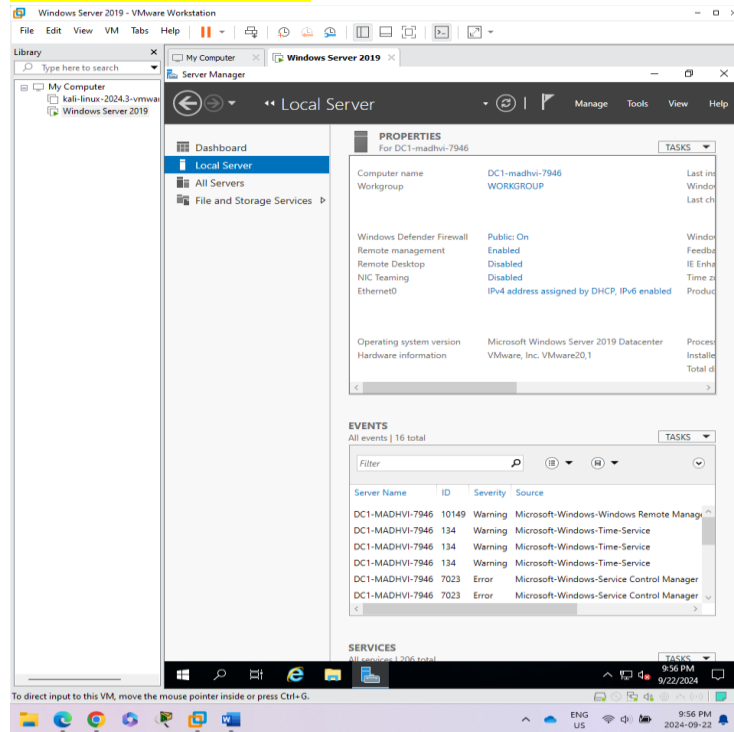
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[Attach Screenshot]



Congratulations! You have successfully installed and configured Windows Server 2019 Datacenter.

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Note:

These steps are typical for a clean installation of Windows Server 2019 on a virtual machine, from loading the installer through to logging into the server and configuring its name. If you are performing the installation on some virtualization software, some minor steps (like loading the ISO) might differ slightly. However, the core steps will remain the same for Windows Server 2019.

When you're done with the first installation, use the snapshot button  on VMware to take a snapshot of your Windows 2019 server virtual machine (This can help you save a clean install of your installation just in a case you receive an error and want to go back to a previous version.) and name it something like (My first snapshot after Installation)

Things to Explore:

You are welcome to explore beyond the mandatory requirements if you wish.

General submission requirements

- Include an opening comment with your full name, date, and a short description.
- Submit your document to DC Connect as firstname_INFT1103.LAB1

Marks distribution:

Questions:

- a) Question 1 & 2 = 35 marks

Lab 1:

- a) 13 Screenshots each 1 mark = 65 marks

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Information Systems Security – Computers and Networking
INFT1103 Operating System Security

Lab 2: Installation of windows 10 and Joining a Domain Controller

Name – Student ID	Course Code	Weight
Madhvi Lashkaree (100997946)	INFT1103	5%

Lab Objectives

Upon completion of this homework, you will be able to perform the following:

- Install Windows 10 Virtual Machine (VM).
- Configure Windows 10 VM and Operating System.
- Configure IP address for windows 10 and dc
- Join the client (Windows 10 VM) to domain controller
- Create user on Active directory

Lab Materials

- **Please name your Lab workbook titled “Lab 2 - <name> <Student ID>”.**
 - Please include all relevant screenshots on this workbook
- Install Windows 10 VM.
- Enter your name and student ID above
- You will need your windows 2016 server installed from previous assignment

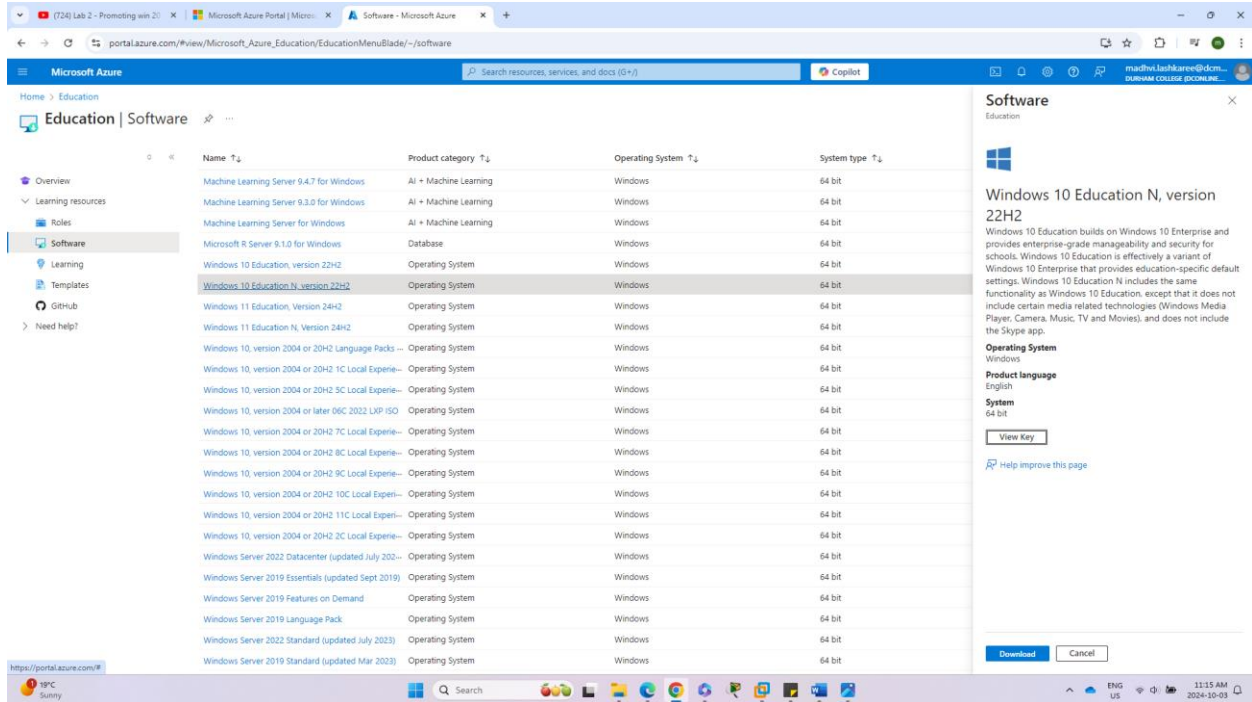
Lab Requirements

- VMware workstation 16 pro
- Windows 2016 server
- Windows 10

Part1: Install Windows 10 VM

1. Go to <https://portal.azure.com/> (same place we downloaded our windows 2016 server from)

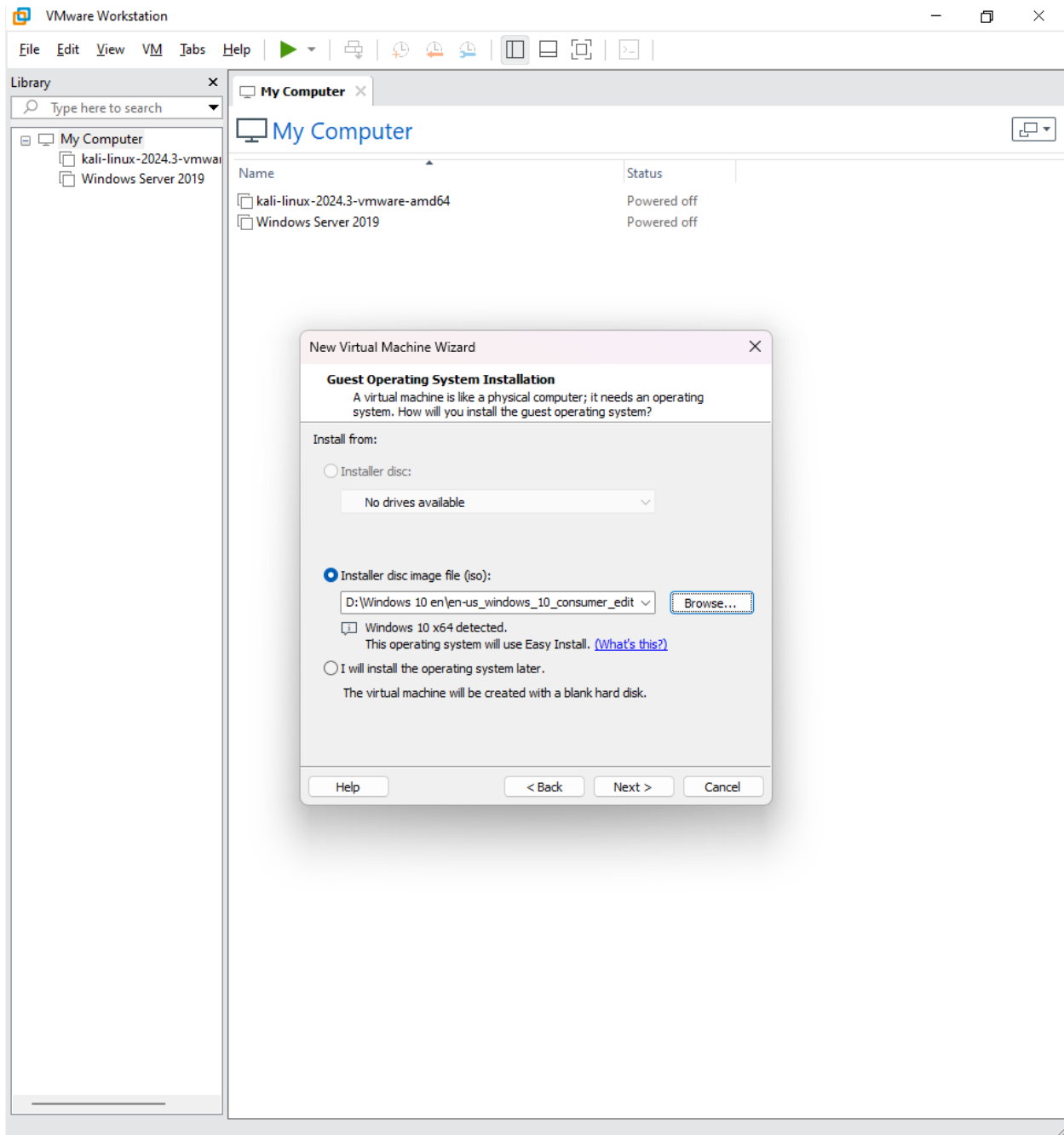
[Attach Screenshot]



2. Grab a copy of Windows 10 Education, version 21H2 – DVD or Windows 10 Education N, version 21H2 – DVD. Make sure to copy the key.

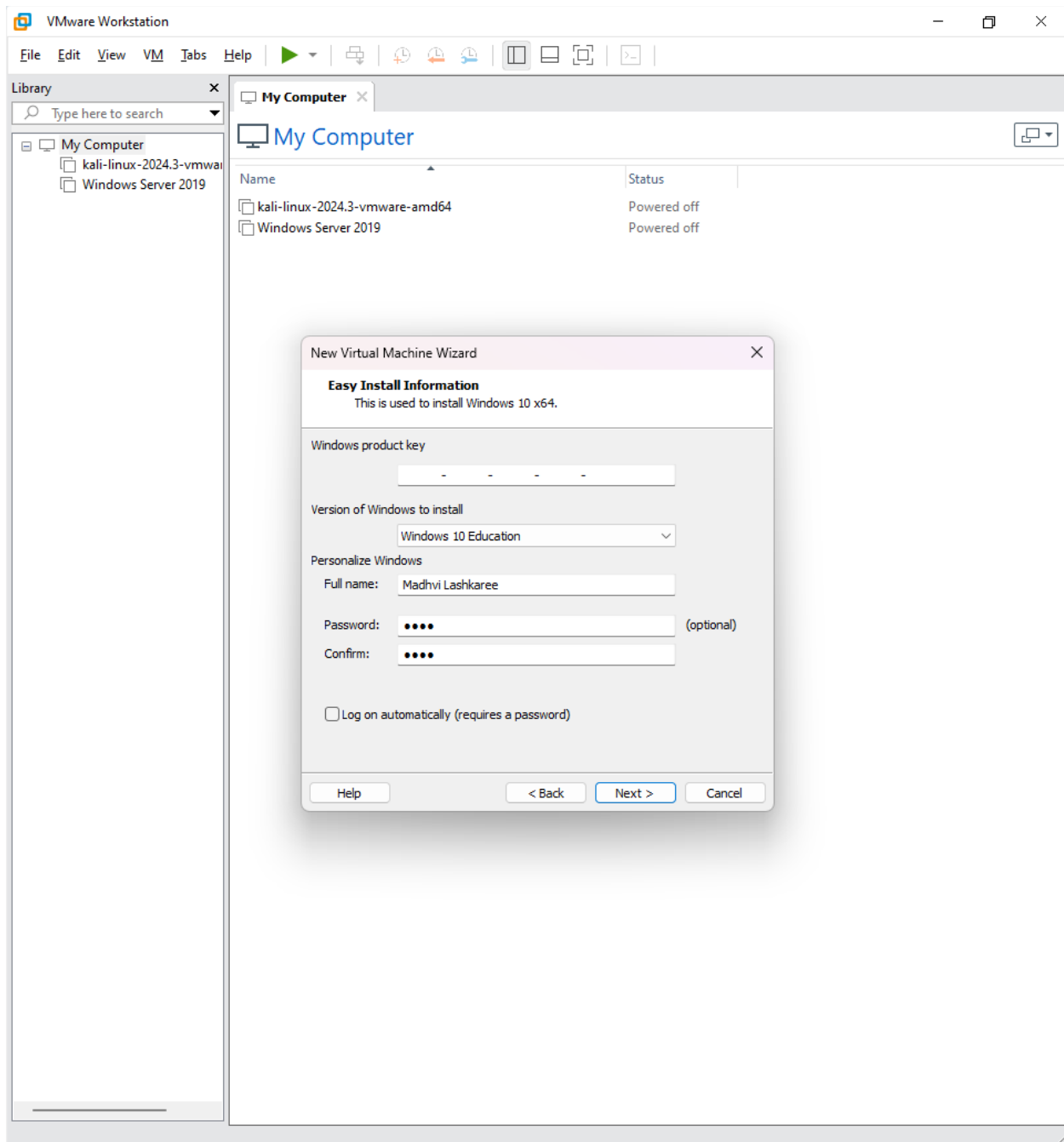
3. Go over to your VMware to install your windows 10. Click on file > New virtual machine. Select windows 10 from the directory it is stored and click Next

[Attach Screenshot]



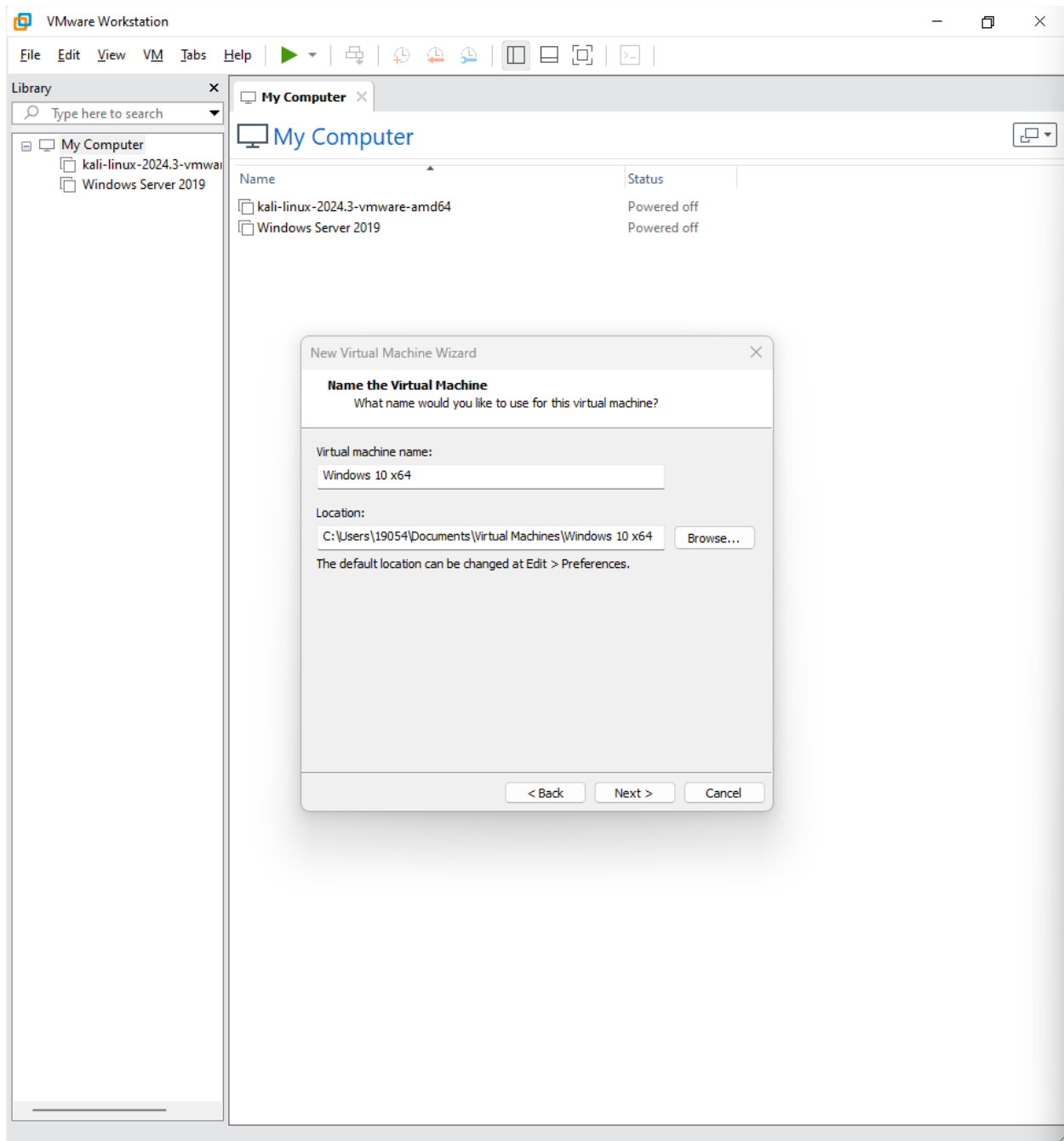
4. Input your license key you got from the Azure website, Input your full name and password. Then click next

[Attach Screenshot]



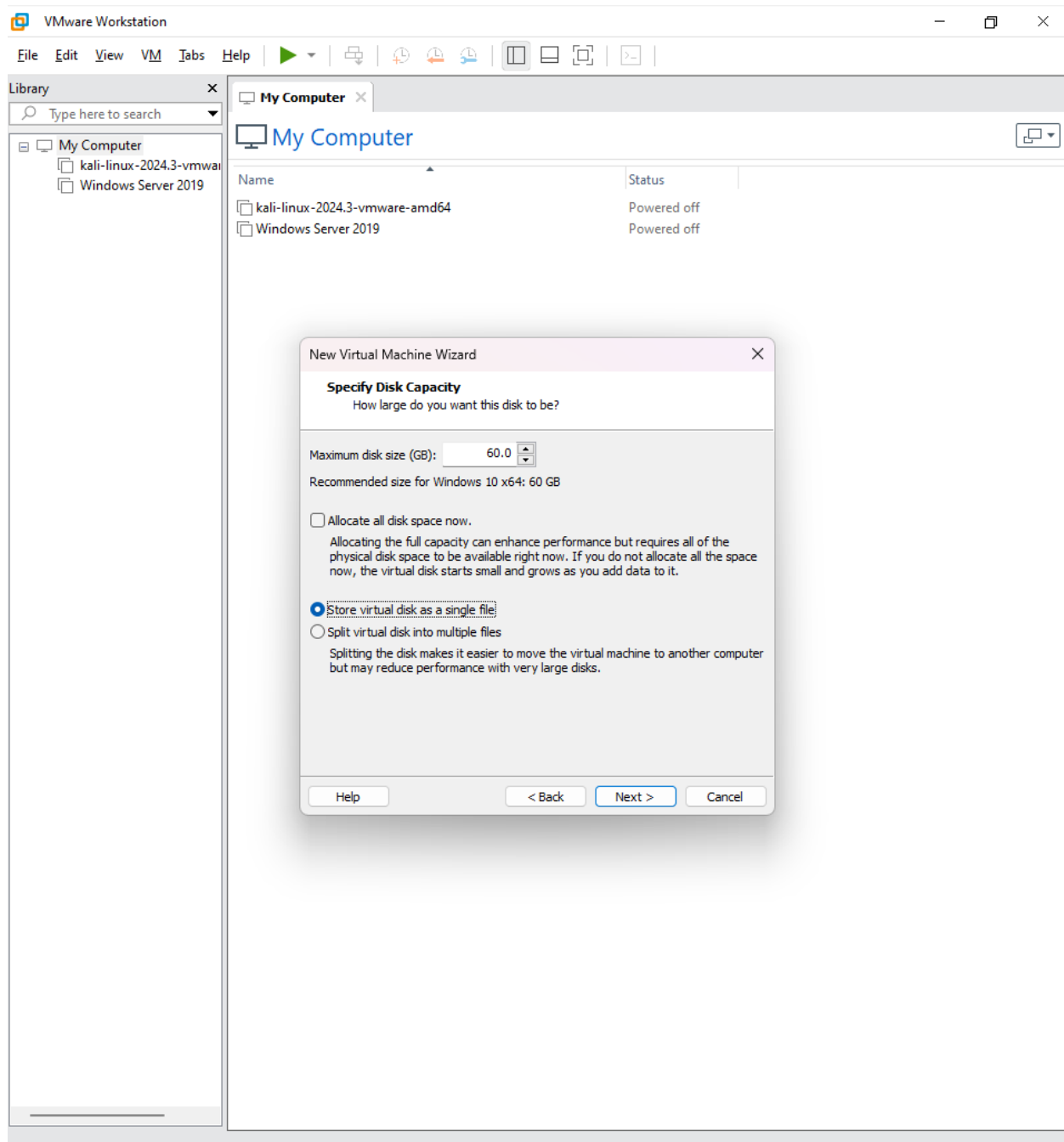
5. Select location to save your virtual machine. For me, I would be storing this in my drive C (you can store this on an external drive)

[Attach Screenshot]



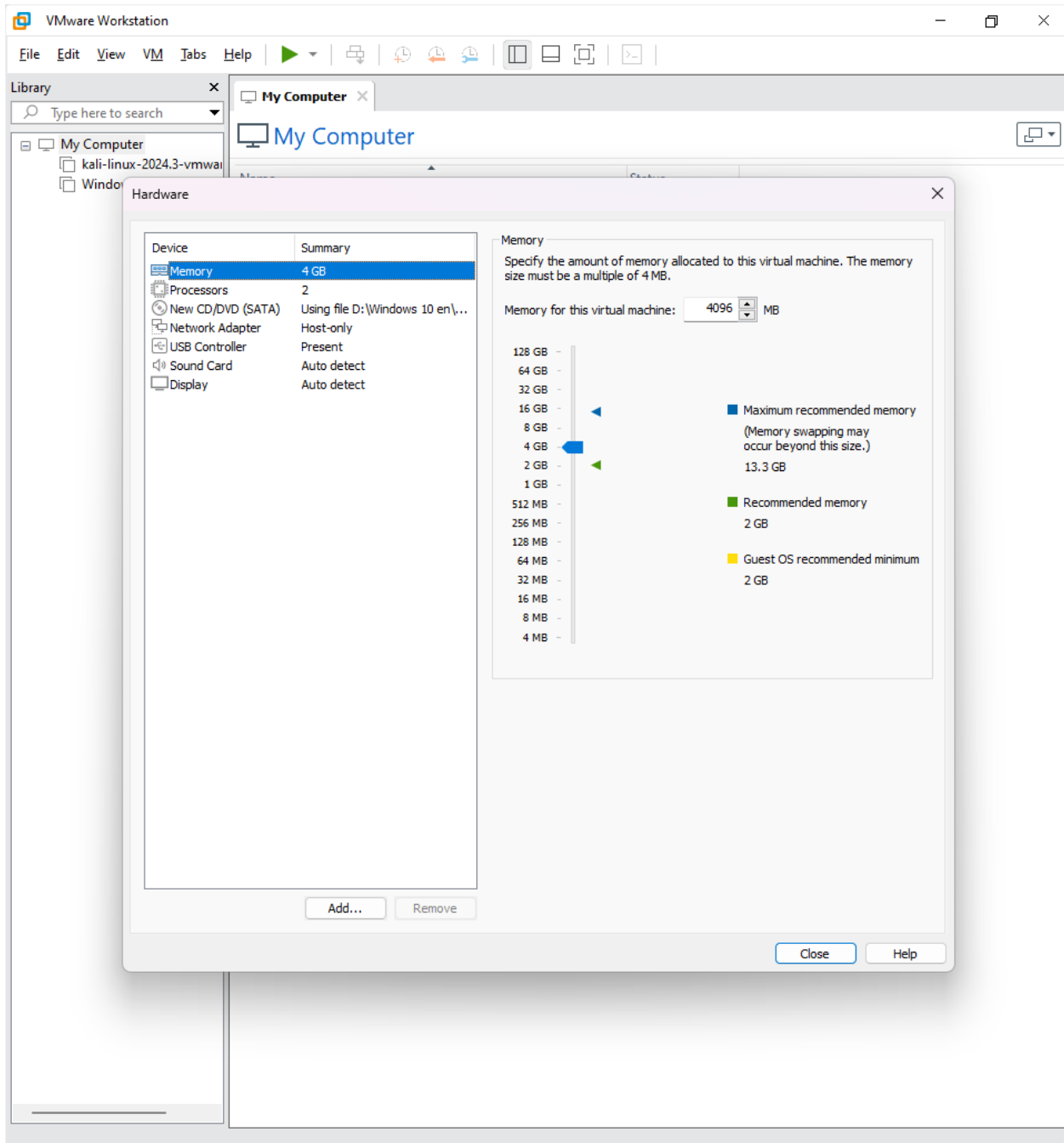
6. Set your drive disk as the default 60gig and select “store virtual disk as a single file” and click Next

[Attach Screenshot]

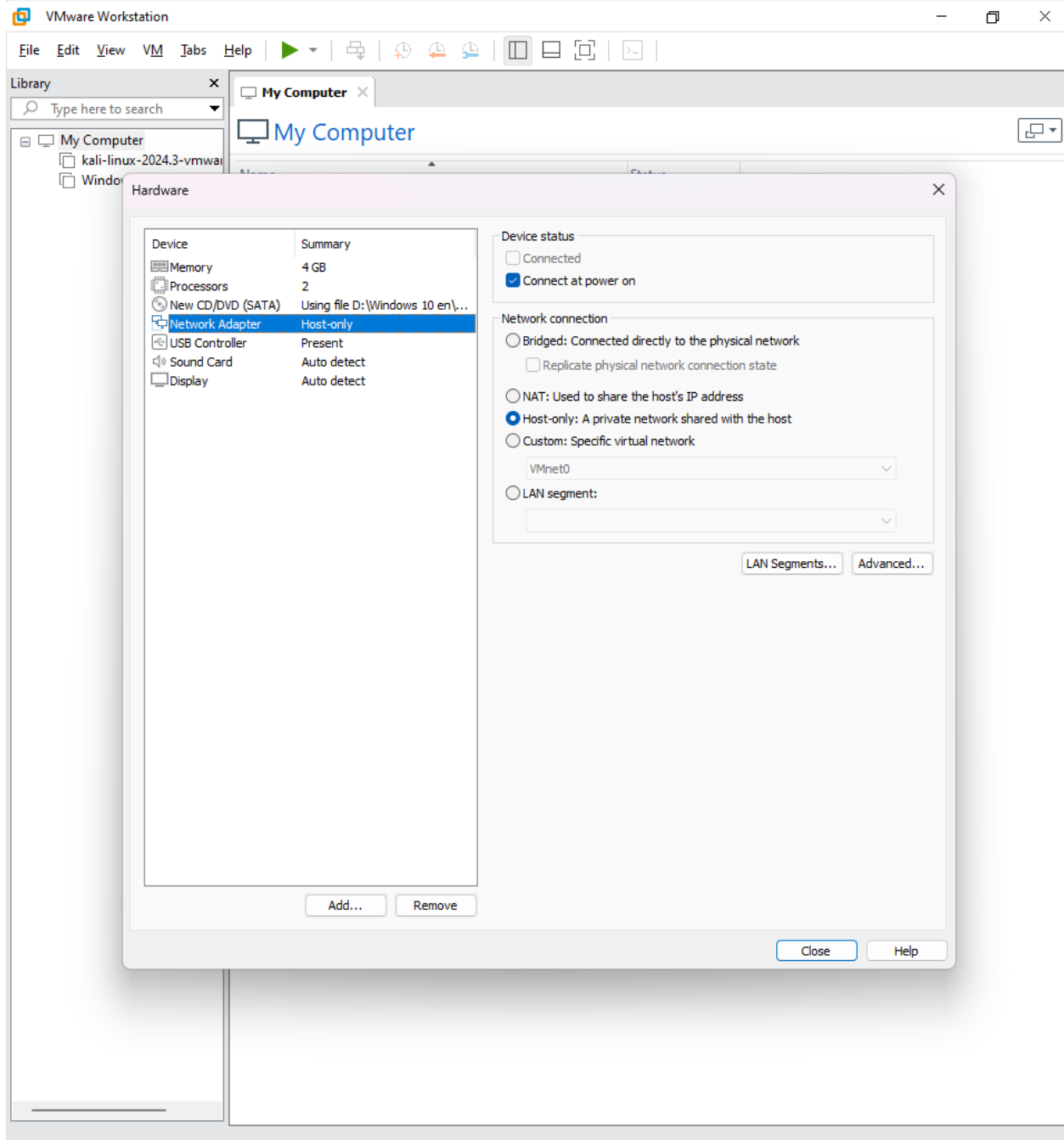


7. Click on customize button, remove printer, set your network adapter to host only and increase your RAM to 4gig (you could choose to add more but 4gig should be enough for our VM to run properly). After making these changes, click on close and finish. Then power on your windows 10 VM.

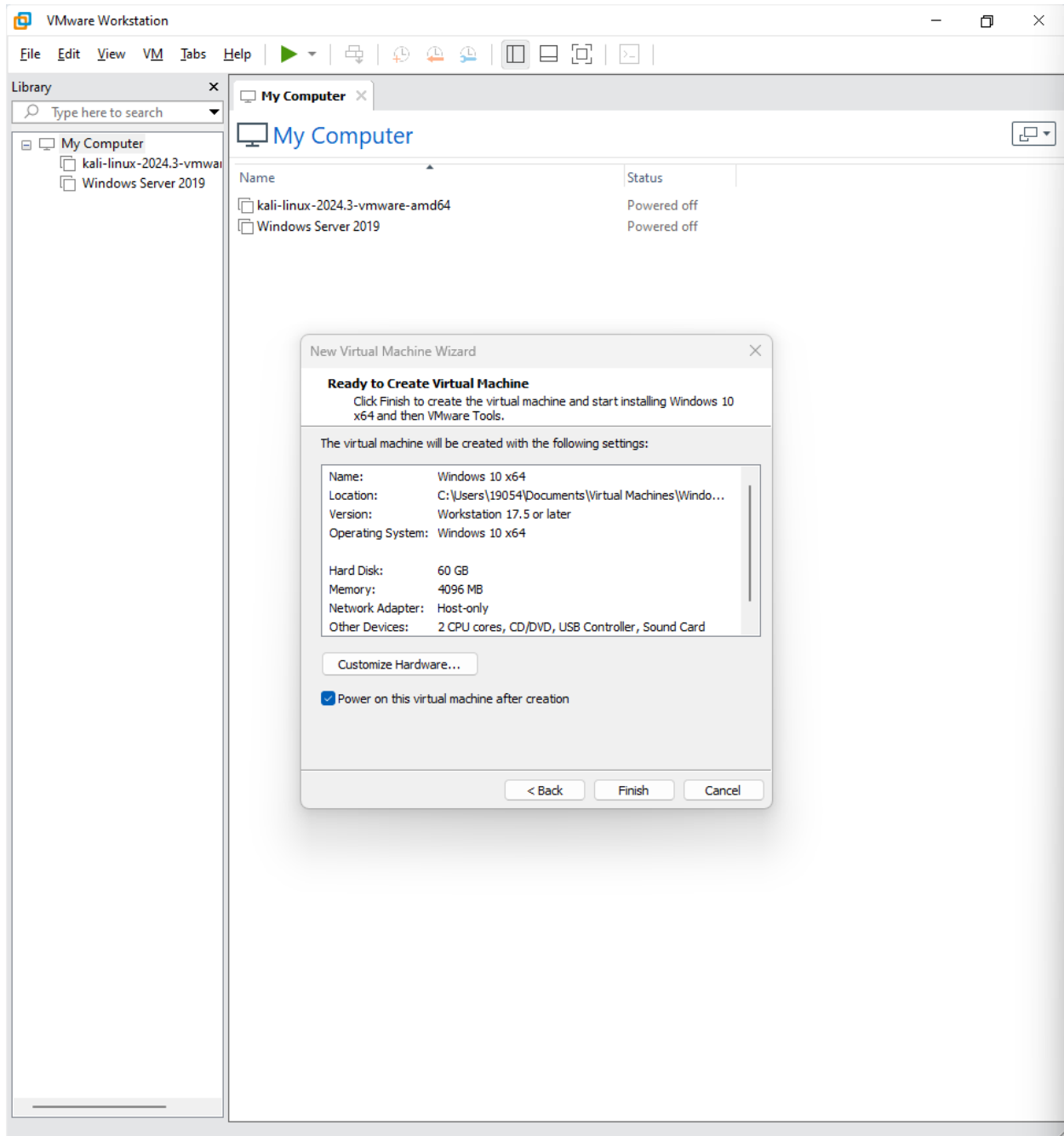
[Attach Screenshot]



[Attach Screenshot]

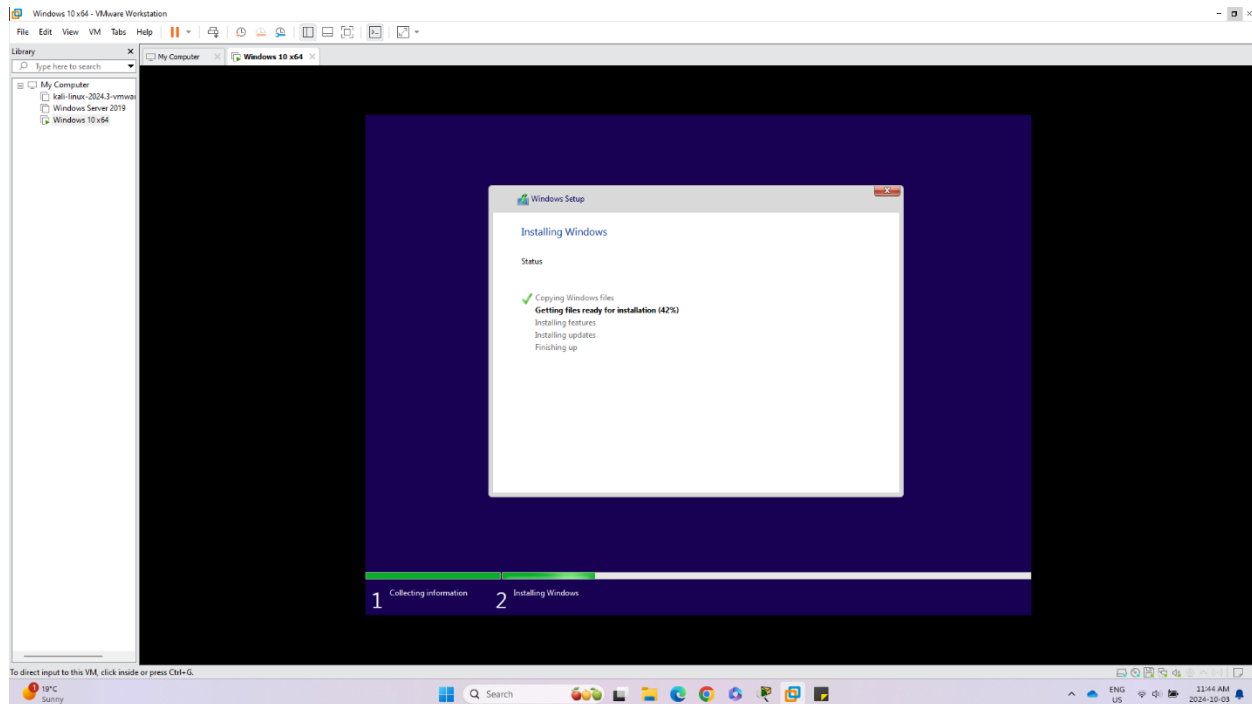


[Attach Screenshot]



8. Once powered on, your windows 10 starts installing immediately, this would take some time. so, you can stretch your legs, grab a coffee while this installs

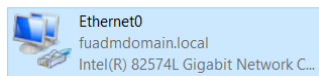
[Attach Screenshot]



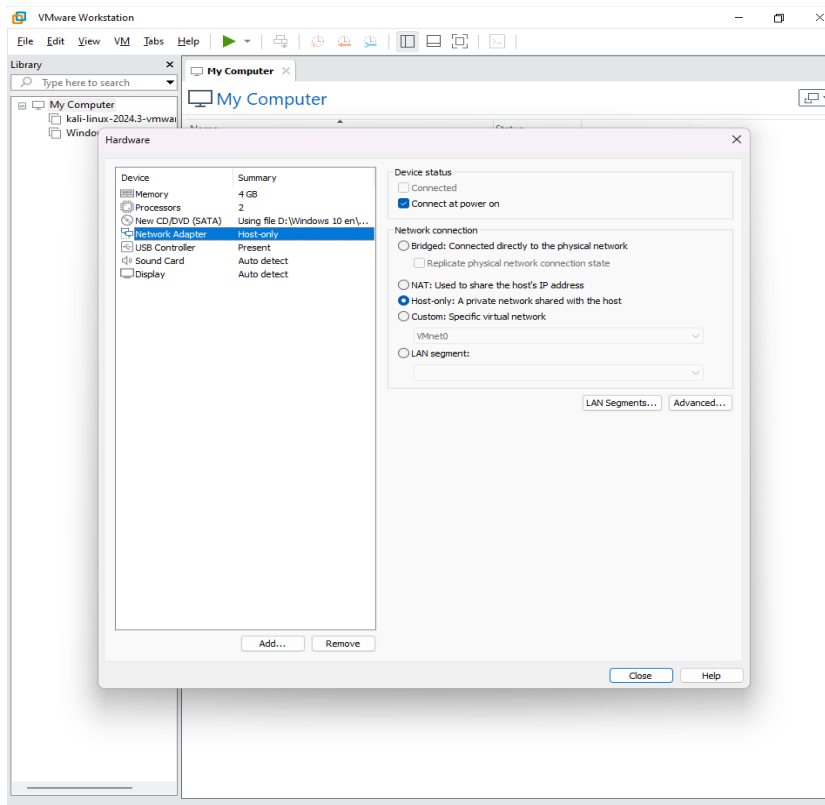
Part 2: We will be linking this workstation to our domain controller

1. Go over to your Domain controller (DC), change the adapter settings to host only and change your IP to a static IP 10.1.1.1 and sent subnet mask to /24. (Note: for a /24 the subnet mask would be 255.255.255.0 as seen in the screenshot)

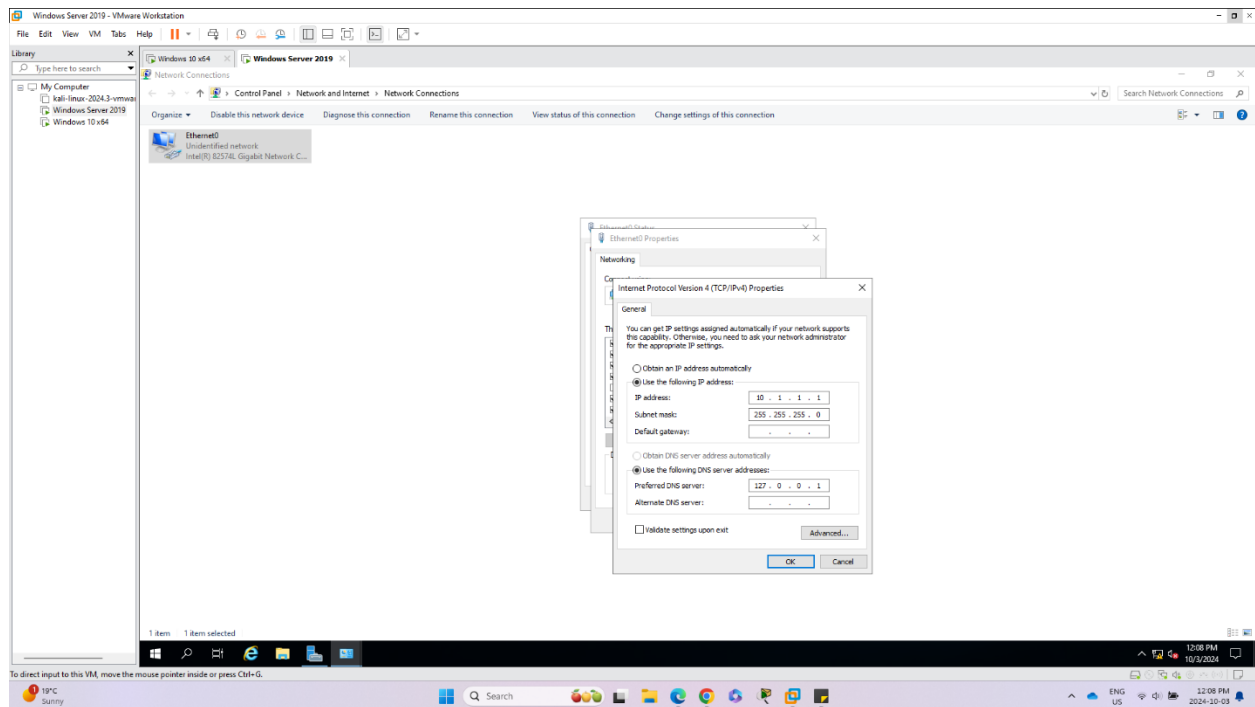
This would be done on your VM by going to Control Panel\Network and Internet\Network and Sharing Center then click on change adapter settings and right click on Ethernet0 > properties > Internet Protocol Version 4 (TCP/IPv4)



[Attach Screenshot]

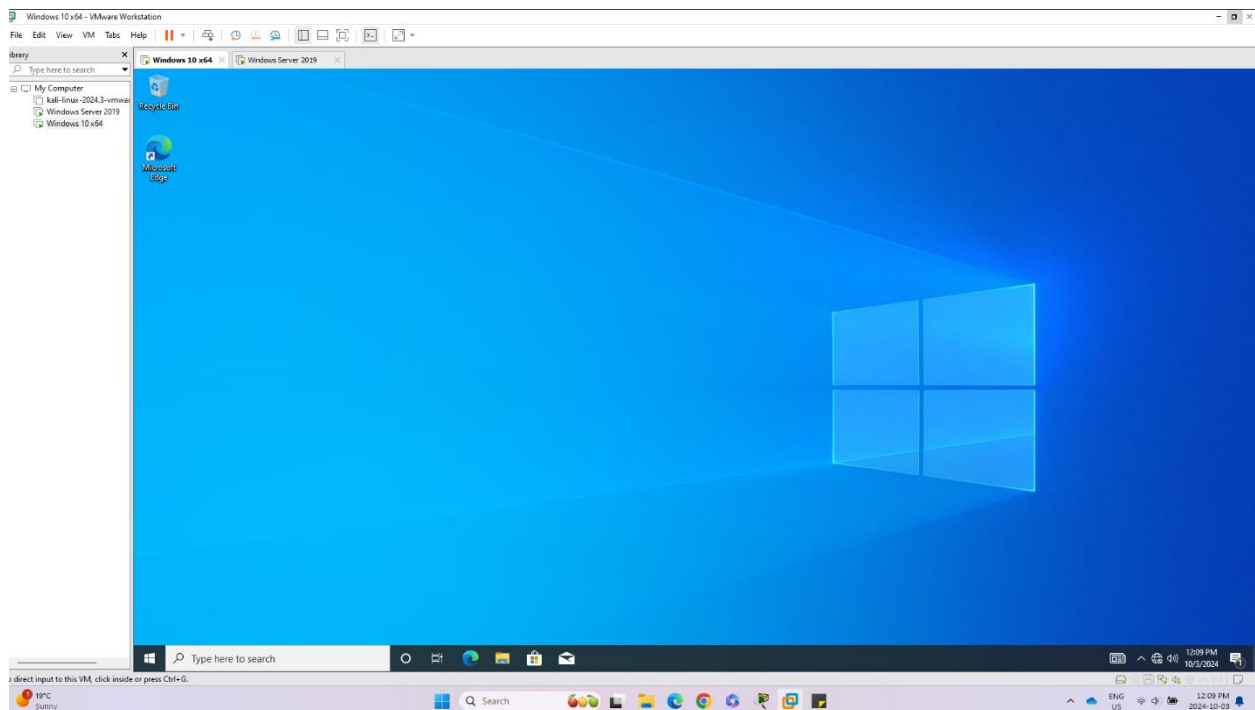


[Attach Screenshot]



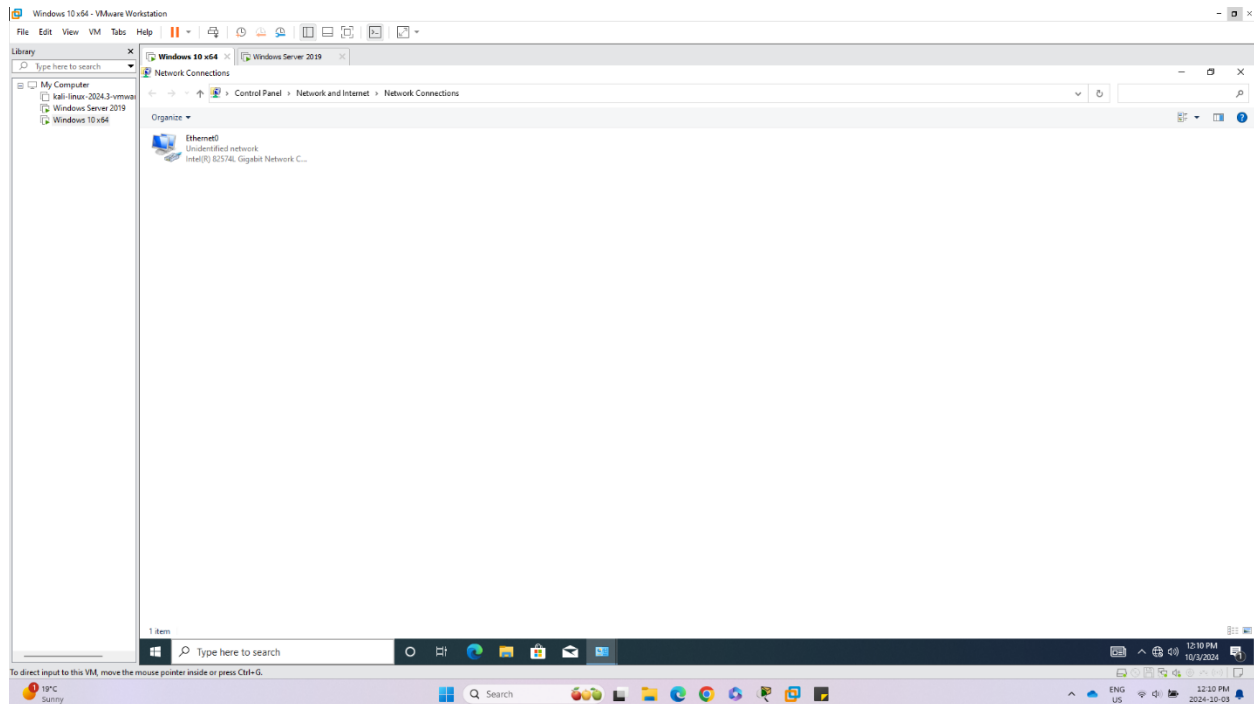
2. Open your newly installed windows 10 to login

[Attach Screenshot]



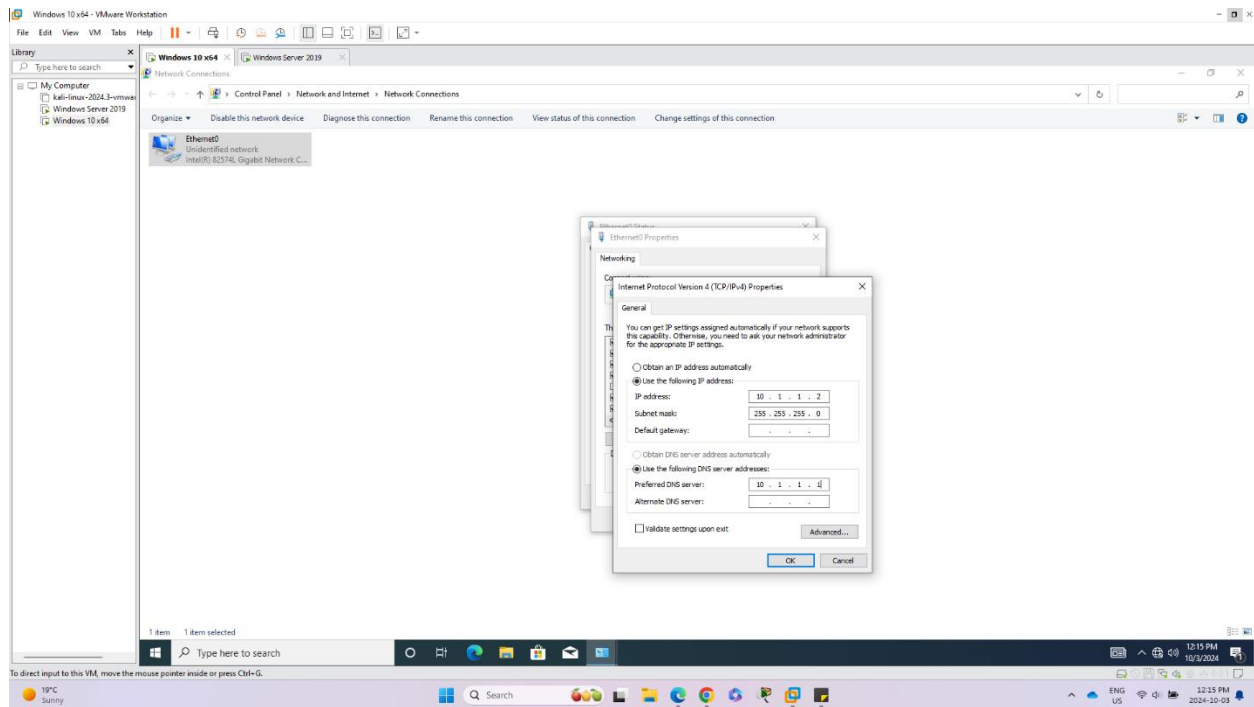
3. Go over to control panel of your windows 10 vm, network and sharing center, change adapter settings and select Ethernet0

[Attach Screenshot]



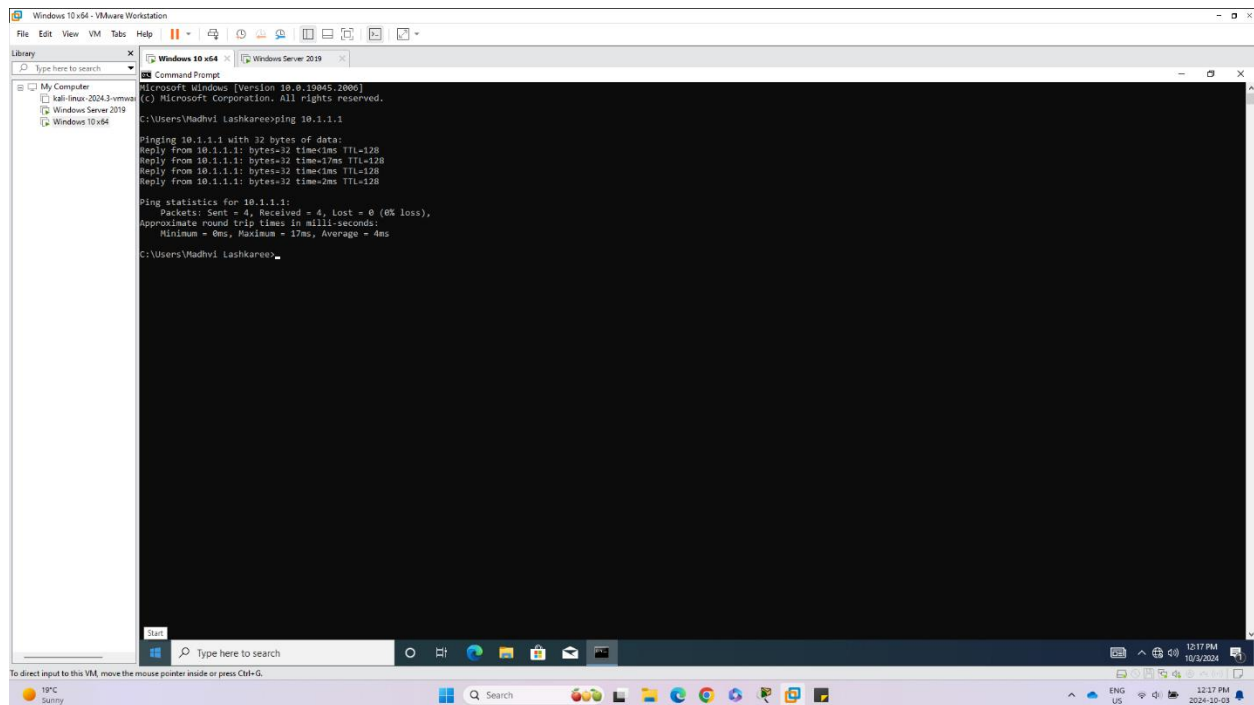
4. Right click, select properties > Internet protocol version 4 (TCP/IPv4) and use a static IP Address of 10.1.1.2, /24 subnet and DNS server would be the Active Directory Server's IP address you set previously (10.1.1.1) and click OK

[Attach Screenshot]



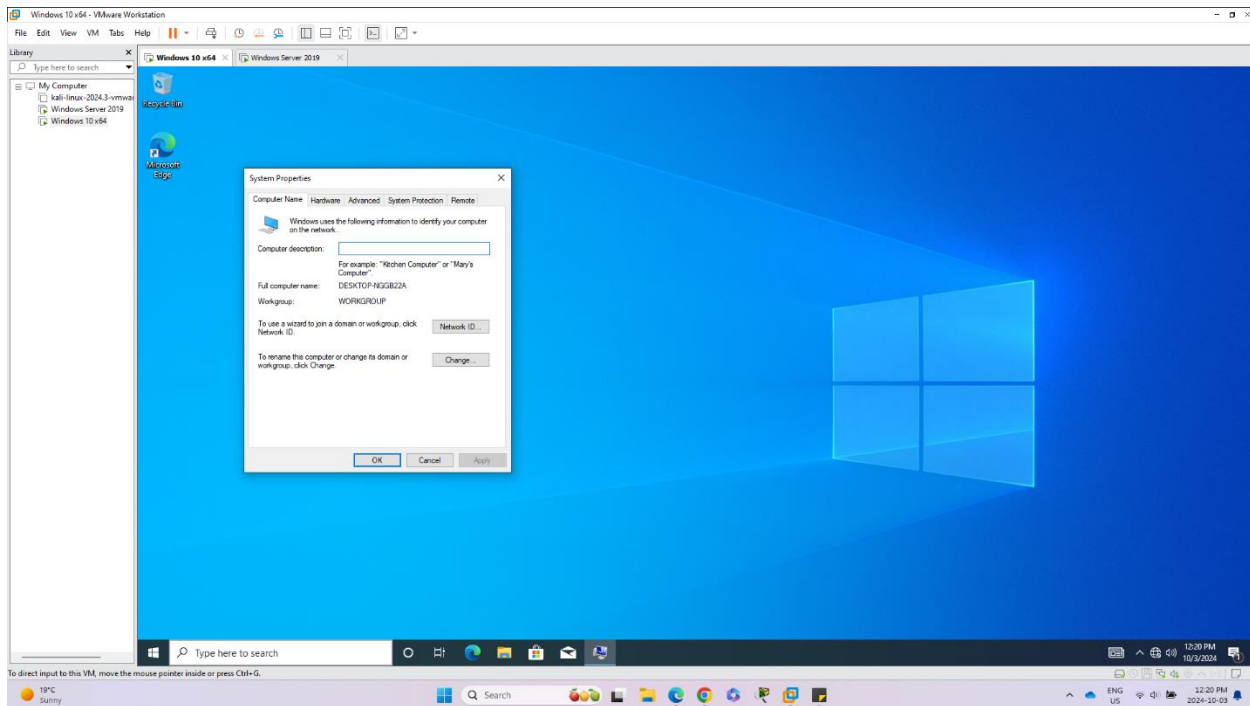
5. Ping your windows server 2016 IP (make sure you can ping your server vm)

[Attach Screenshot]

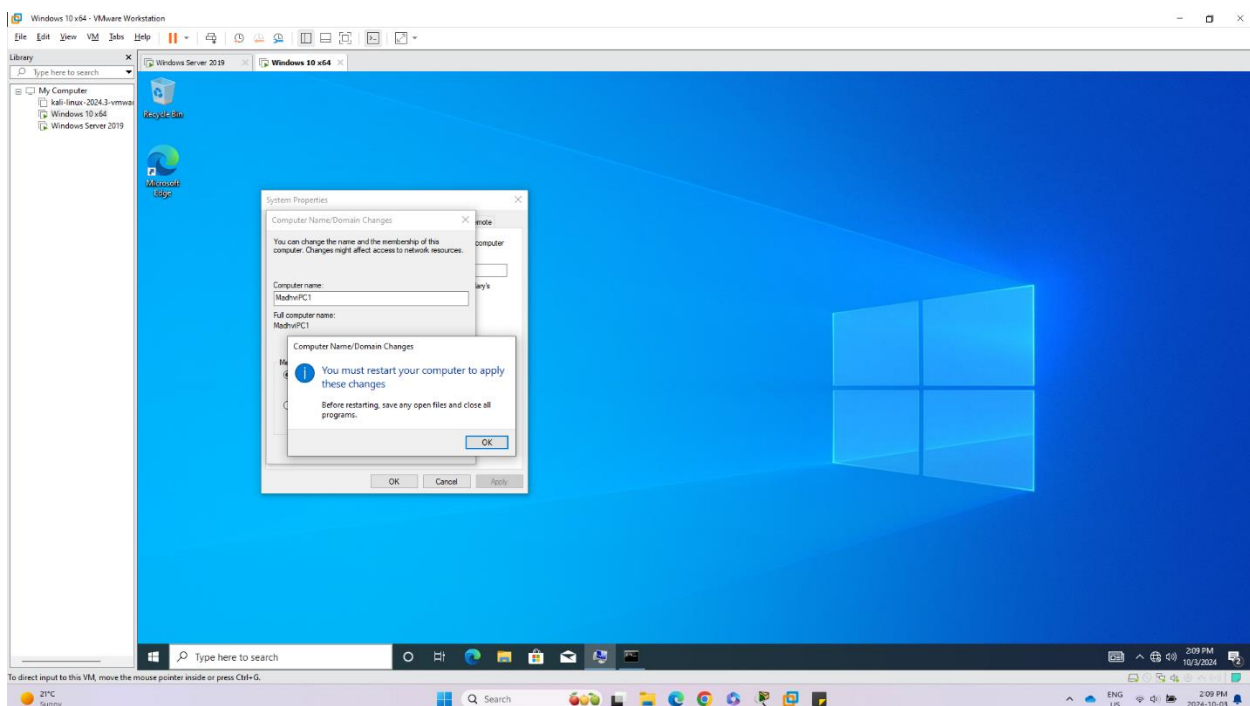


6. Click start, type '**view advanced system setting**' click 'Computer Name' click 'Change' input the domain name into the 'Member of' section and change the computer name to YournamePC1

[Attach Screenshot]

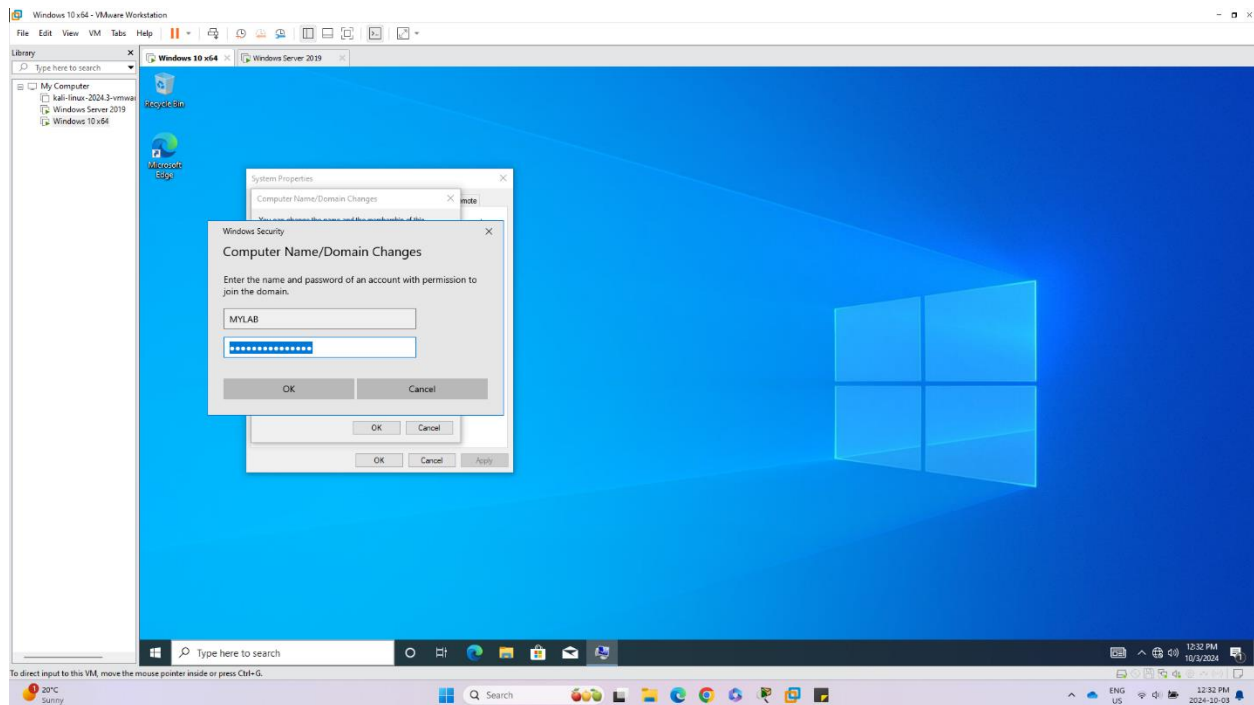


[Attach Screenshot]

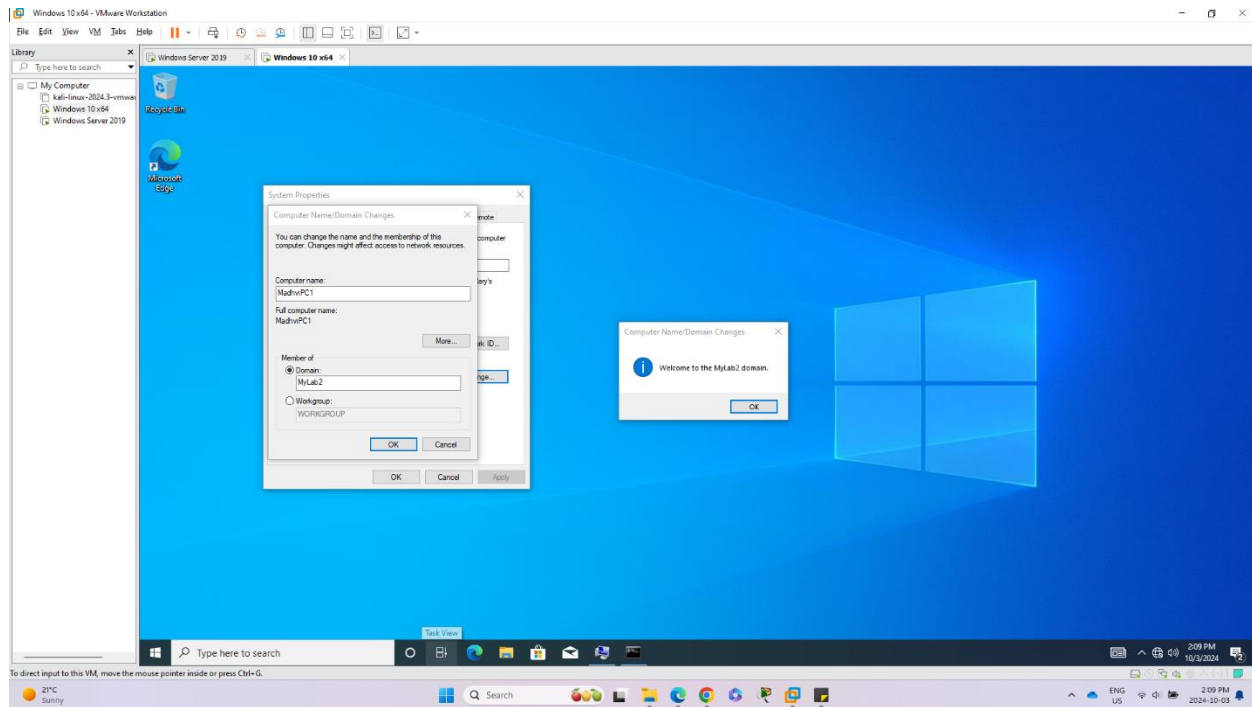


- Click OK and it would prompt you for the domain username and password. Input your domain controller's username and password. Click OK, you should get a message saying "Welcome to the <domain name> domain". You will need to restart your Windows 10 VM after it is a member of the domain.

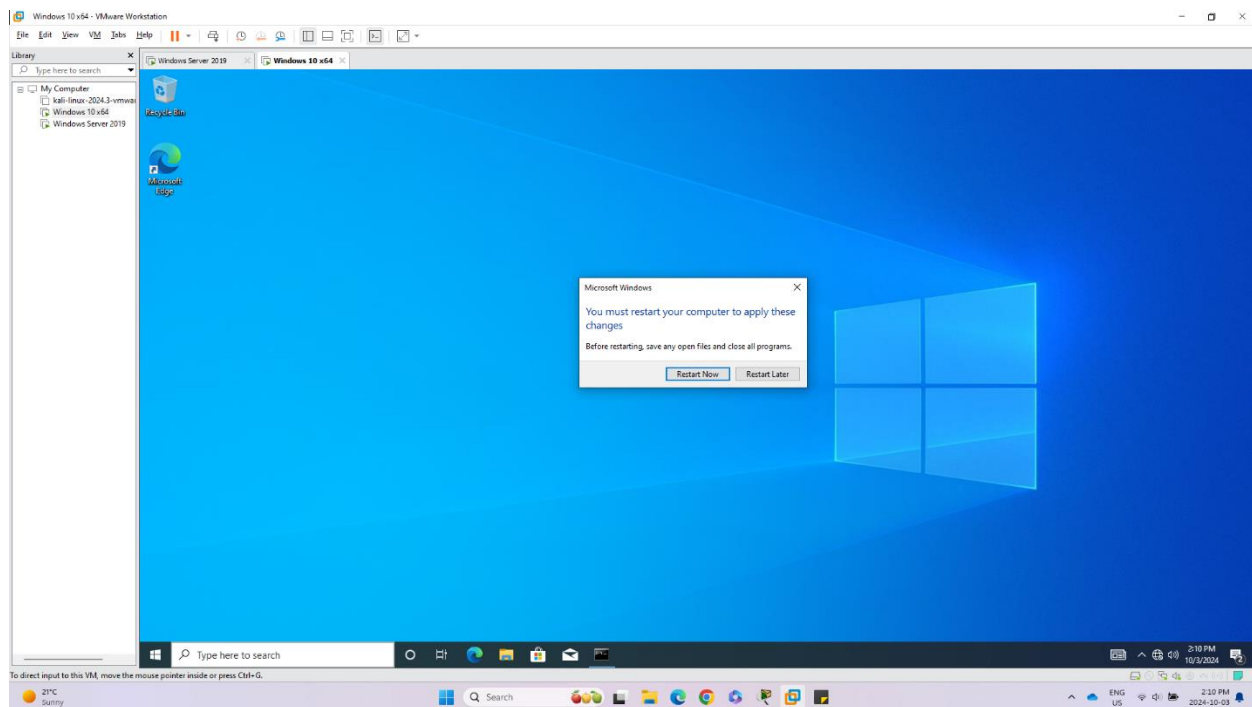
[Attach Screenshot]



[Attach Screenshot]

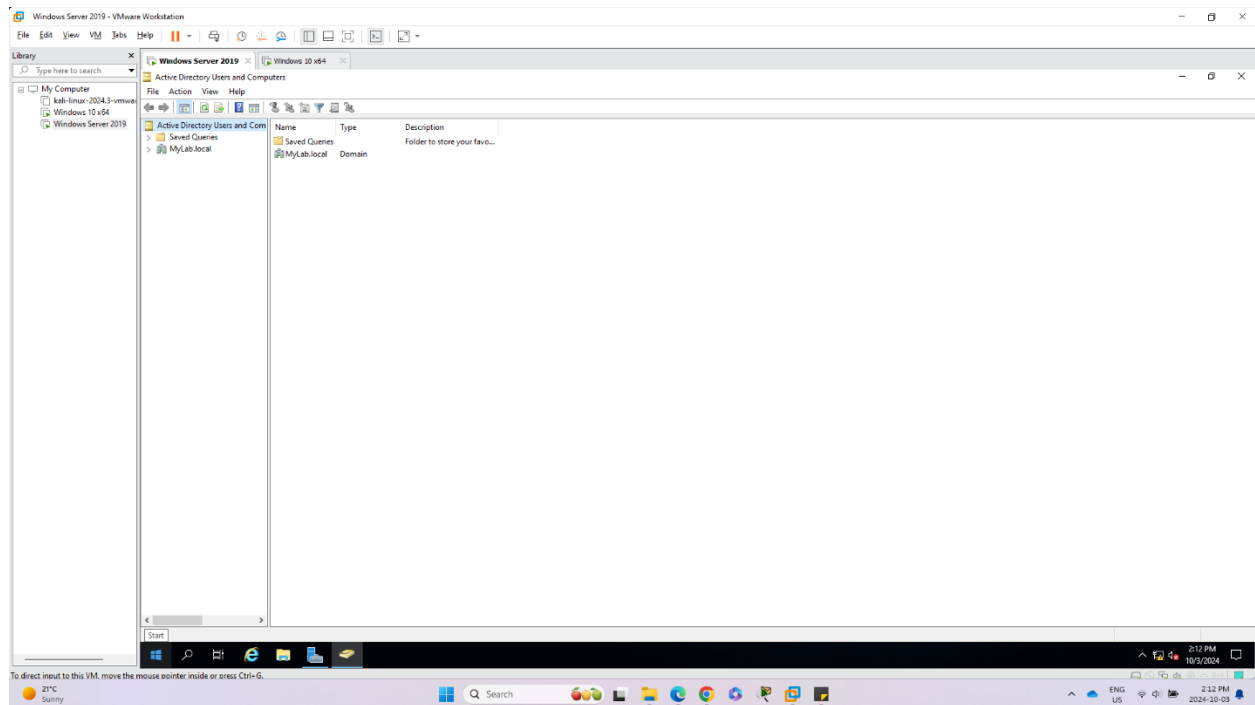


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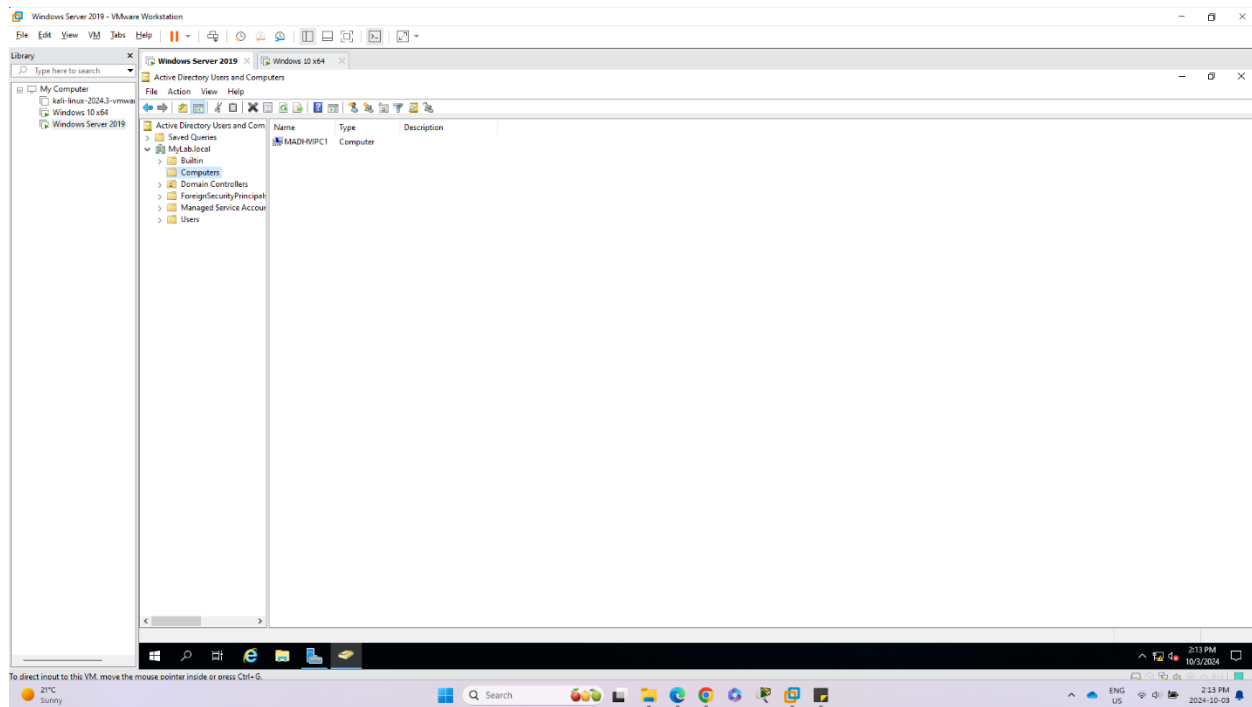
8. Your windows 10 should be a part of the computers added to your domain controller. Go over to your domain controller. Go to start and type Active Directory Users and Computers.

[Attach Screenshot]



- Click on our domain dropdown and click on computers, we should see our windows 10 VM added to the list

[Attach Screenshot]



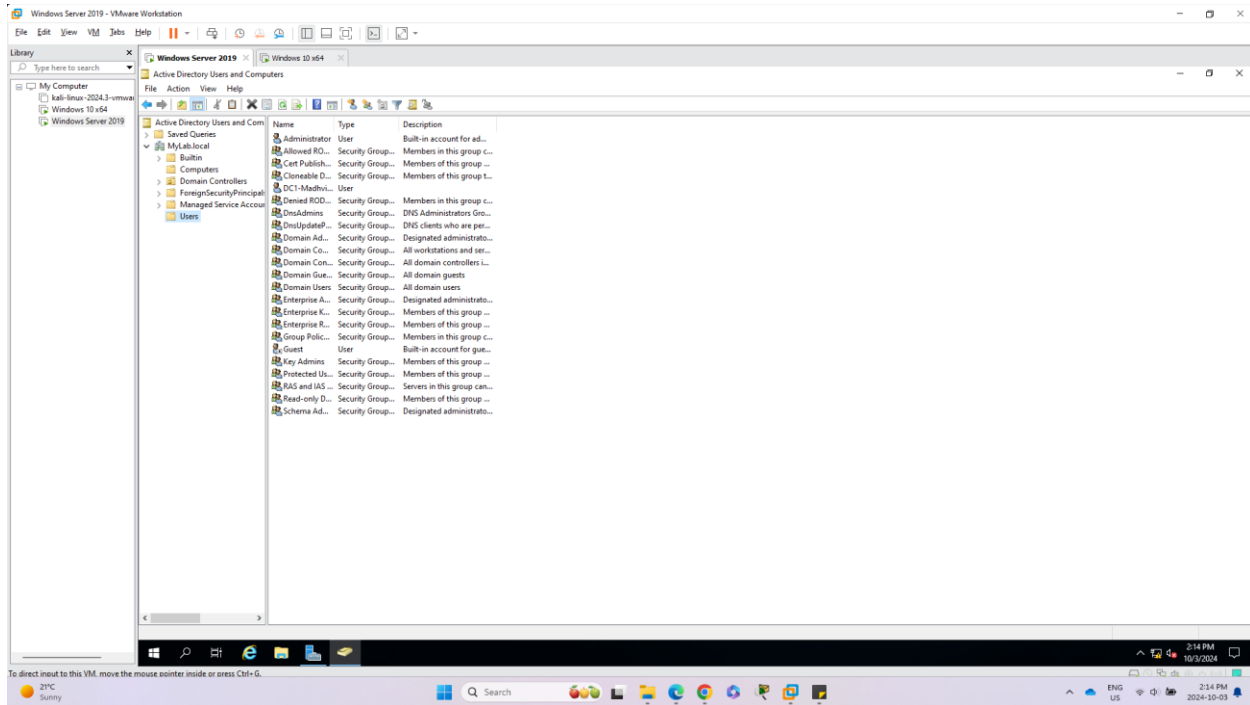
- Good Job you have finally added your windows 10 VM to your domain controller.

Part 3: Create a new user in Active directory

We would be creating a new user in active directory and login into our windows 10 PC using the new user account created.

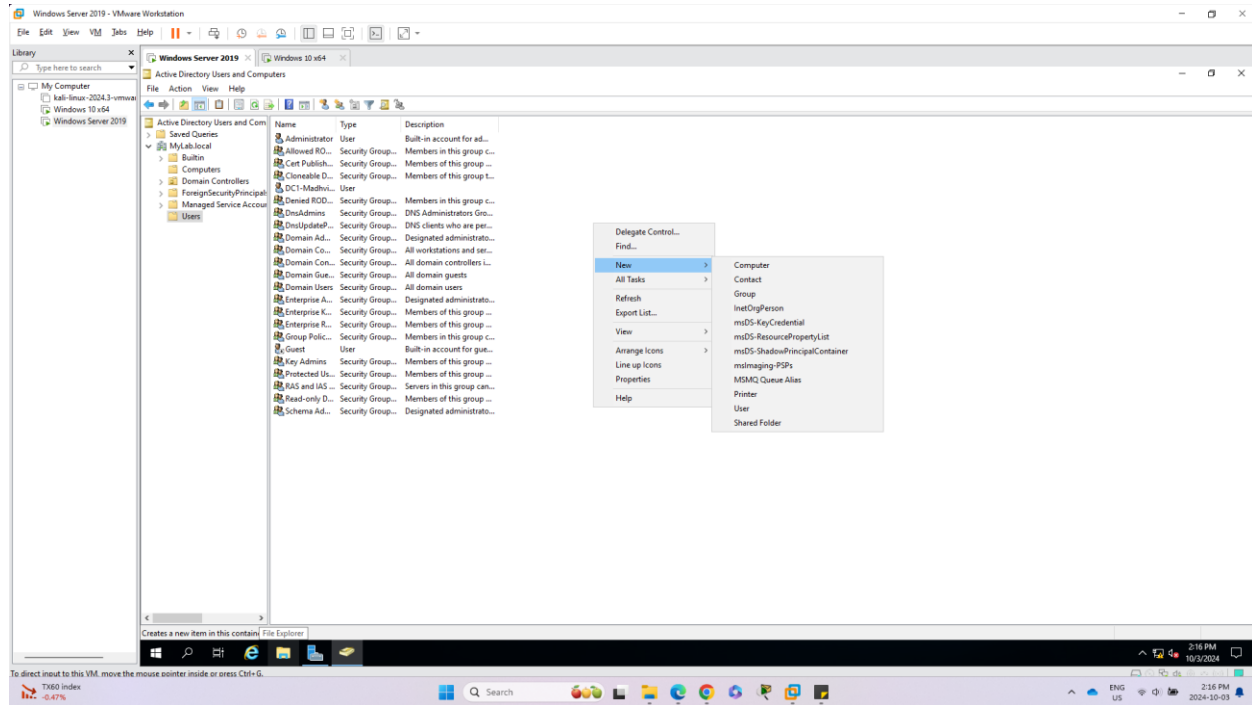
1. Go to your domain controller. Click start and type Active Directory Users and Computers. Click users

[Attach Screenshot]

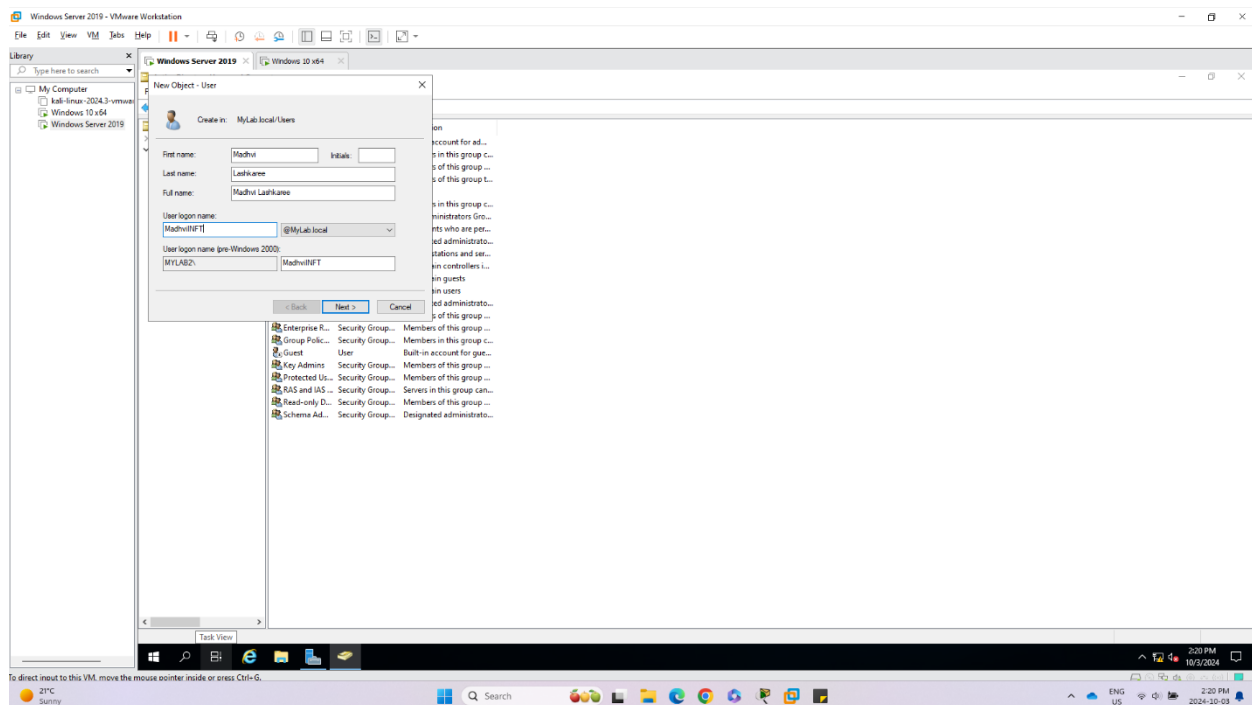


2. Right click, click New > user

[Attach Screenshot]

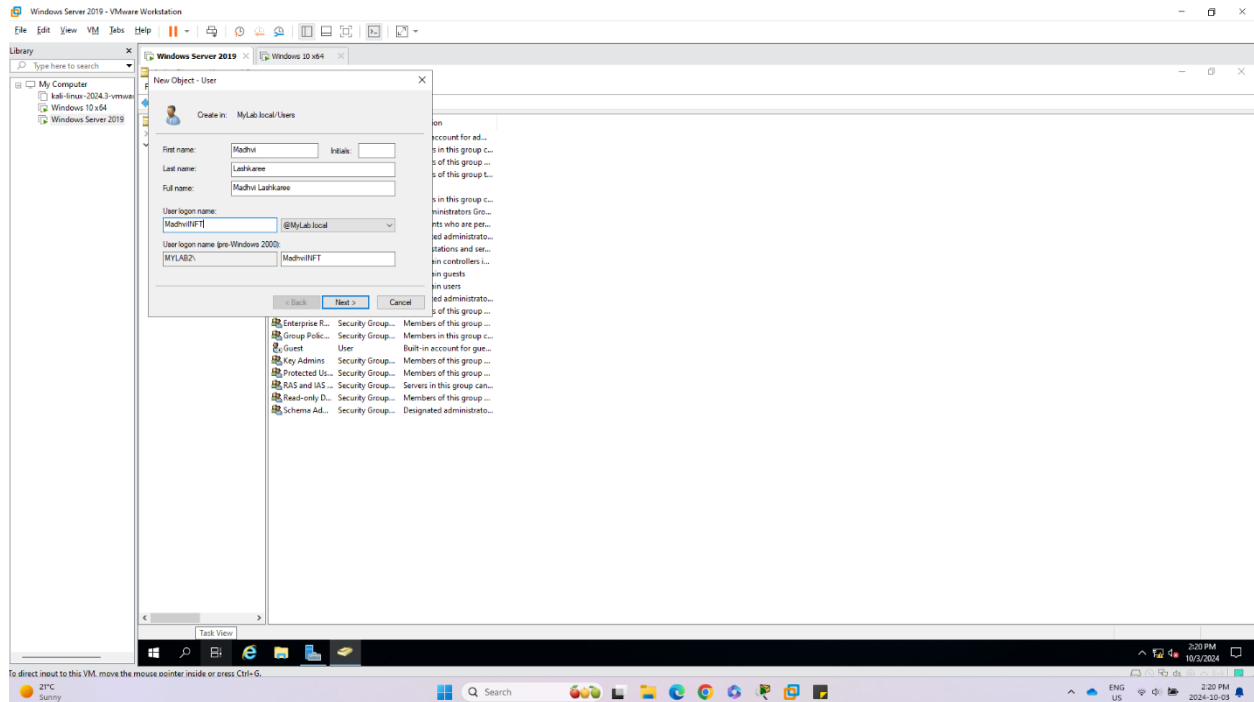


[Attach Screenshot]



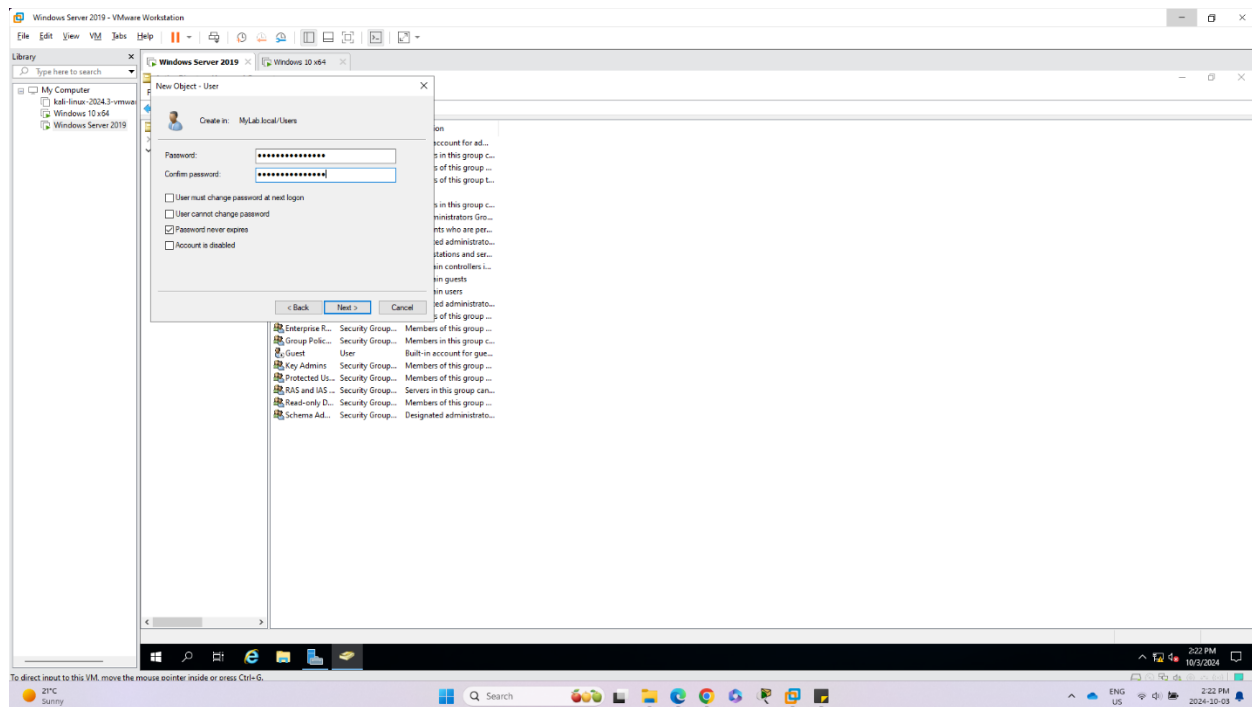
3. Create a new user with the format YournameINFT i.e., fuadINFT (make sure you use the right format for the user logon name)

[Attach Screenshot]



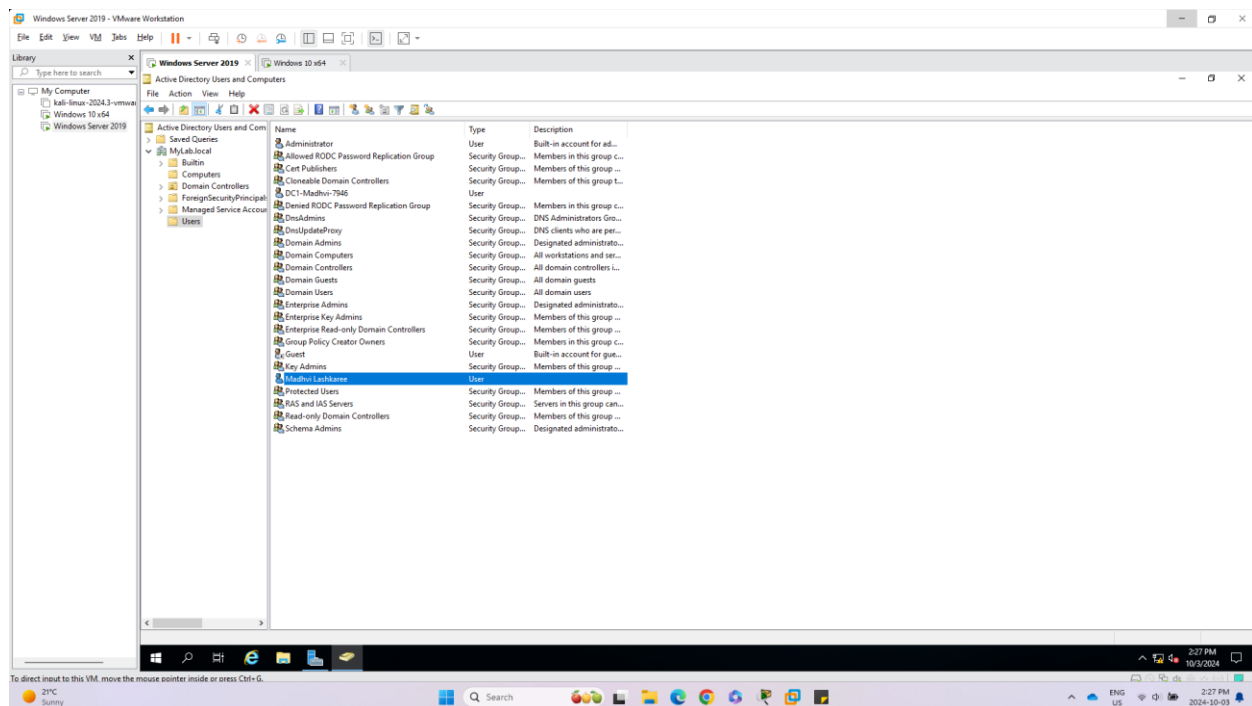
4. Click Next and set a password and set password to “**Password never expires**”.

[Attach Screenshot]



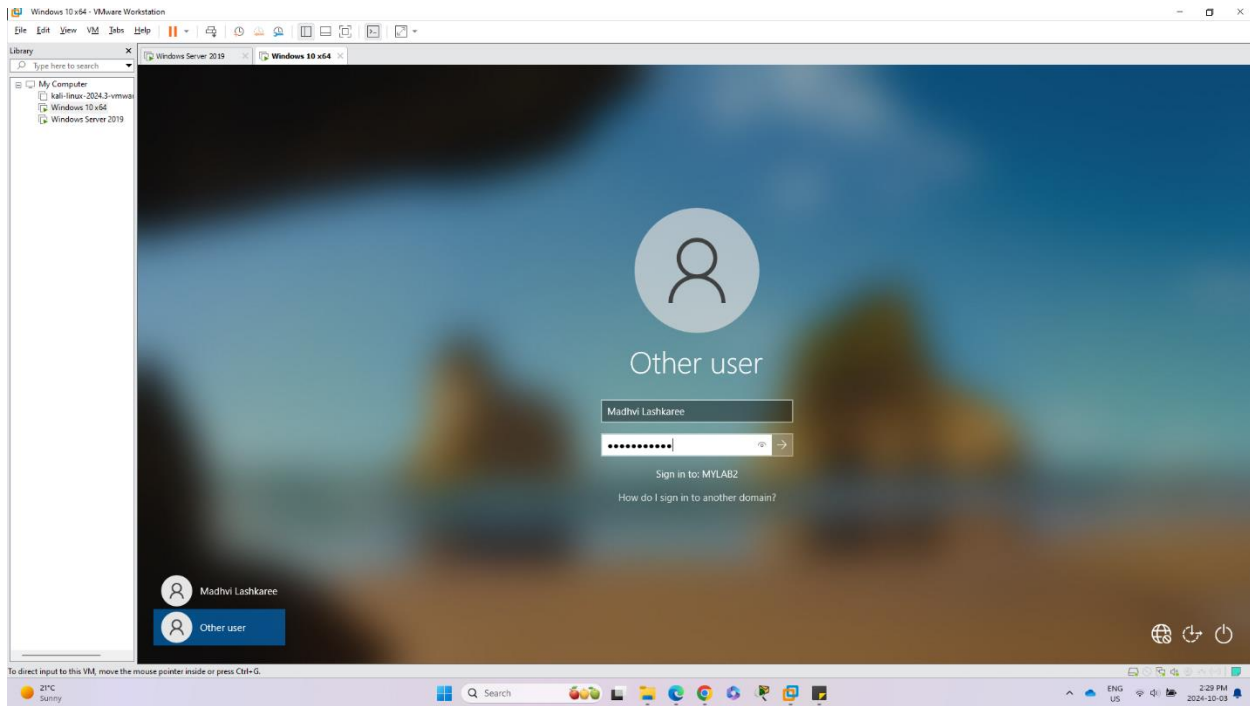
5. Your new user would be added to the users in your domain controller

[Attach Screenshot]



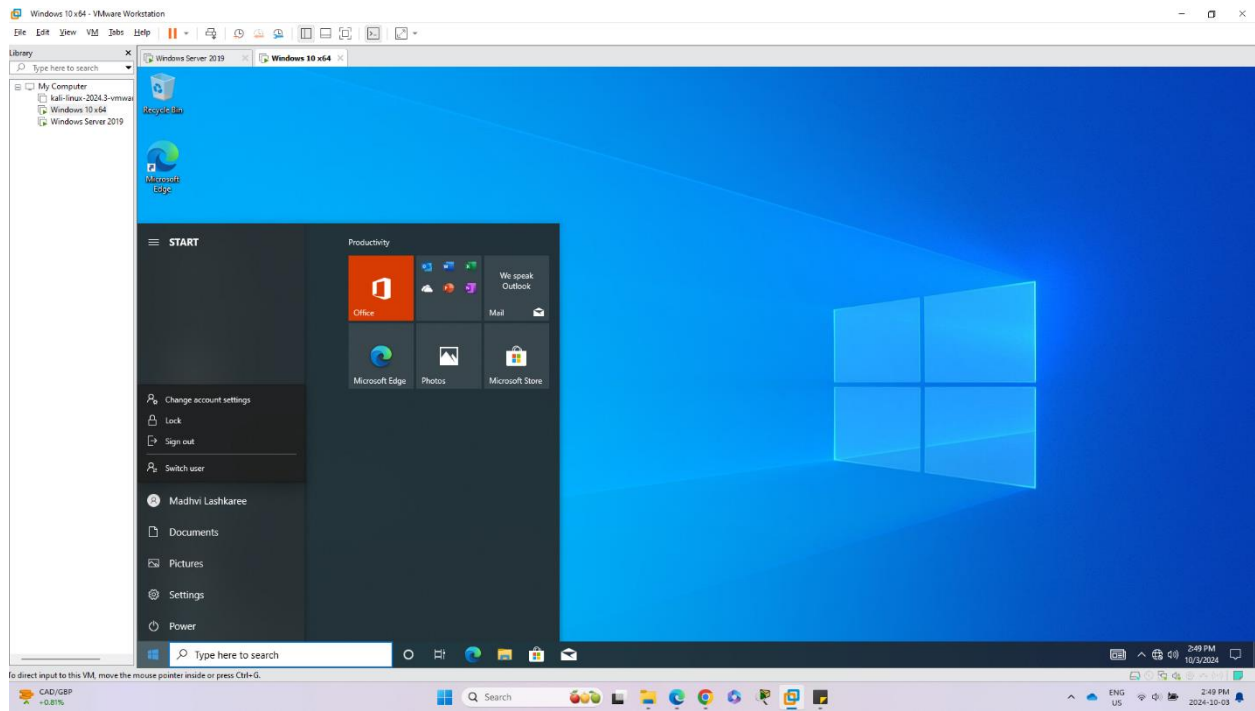
6. Go over to your windows 10 VM to login to your new user

[Attach Screenshot]



7. After logging in, go over to your start menu and take a screenshot showing your username

[Attach Screenshot]



Things to Explore:

You are welcome to explore beyond the mandatory requirements if you wish.

General submission requirements

- Include an opening comment with your full name, date, and a short description.
- **Marks distribution:**
 - a) 30 Screenshots each 1 mark = 30 marks
 - b) Video Demonstration = 5 marks
- **Do not alter the sequence of steps of this document. Do not delete the words [Attach the screenshot] anywhere from the assignment. Keep the numbering same. Paste the screenshots wherever it's being asked- do not use screenshots of professors from lab manual. They are for your reference only and would be considered an Academic Integrity Breach. You should be not even submitting another student's screenshot- Academic Integrity alert.**
- **Submit a video recording (sharing screen) briefly explaining the steps taken to perform the lab. If the video recording stating the above requirements are not submitted, you will be eligible to get marks out of 60% of Lab2 marks.**

*****GOOD LUCK*****